Mathematical Tripos

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Part III

Lie Algebras and their representations

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Contents

1 Introduction **1**

2 Basic Definitions and Examples **3**

3 Basic Representation Theory **6**

4 Representation Theory of $\mathfrak{sl}_2(\mathbb{C})$ **9**

5 Irreducible modules for $\mathfrak{sl}_2$ **13**

6 Tensor products **17**

7 Results about semisimple Lie algebras **19**

8 Jordan decomposition **29**

10 Cartan subalgebras and root space decompositions **33**

11 Root systems **43**

12 Weyl chambers and root bases **47**

13 Facts about the Weyl group **50**

14 Classification of irreducible root systems **51**

9 Brief introduction to inner automorphisms **59**

15 Conjugacy results **60**

16 Weights **61**

17 The PBW theorem **65**

18 Highest weight and Verma modules **70**
18.1 Towards Verma Modules! 72

19 The Weyl character formula **77**

1 Introduction

A Lie group is, fundamentally, a group that is also a smooth manifold, for example:

$$GL_n,\ SL_n,\ SO_n,\ Sp_{2n}$$

Over, for example, some infinite field, such as $\mathbb{C}$.

Example 1.1. $S^1 = \{z \in \mathbb{C} \mid |z| = 1\}$ is the prototypical example of a Lie group.

Let G be a Lie group. The Lie Algebra of G is the tangent space of G at the identity, e, of G. That is to say:

$$\mathfrak{g} = T_e G.$$

$\mathfrak{g}$ is a vector space, but also has additional structure.

By taking a differential, we can turn the conjugation map:

$$\begin{aligned} G &\to \operatorname{Aut} G \\ g &\mapsto g(\,\cdot\,)g^{-1} \end{aligned}$$

where G is a Lie group, into a map ad (for adjoint):

$$\operatorname{ad} : \mathfrak{g} \to \operatorname{End} \mathfrak{g}.$$

This gives a bilinear map:

$$\begin{aligned} [\,\cdot\,,\,\cdot\,] : \mathfrak{g} \times \mathfrak{g} &\to \mathfrak{g} \\ [x, y] &\mapsto \operatorname{ad}(x)(y) \end{aligned}$$

Example 1.2. If $G = GL_n(\mathbb{R})$, then

$$\mathfrak{g} = \mathfrak{gl}_n(\mathbb{R}) = M_n(\mathbb{R})$$

$$[x, y] = xy - yx$$

So far, so good, so what?

(a) How do Lie algebras tell us about the structure of G?

- We'll define the root system of G; this tells us about commutator relations in G (See Carter's book for more)
- We'll define the Weyl group of $\mathfrak{g}$, e.g. the Weyl group of $GL_n(\mathbb{C})$ is $\cong S_n$. This is basically to do with the fact that there is an embedding

$$S_n \hookrightarrow GL_n(\mathbb{C})$$

which is obtained via permutation matrices acting on a basis of $\mathbb{C}^n$.

Let $\mathcal{B} = \{\text{upper triangular matrices in } GL_n(\mathbb{C})\}$. Then $\exists$ Bruhat decomposition, which is:

$$GL_n(\mathbb{C}) = \bigsqcup_{w \in S_n} \mathcal{B} w \mathcal{B}$$

(i.e. the disjoint union of double cosets of $\mathcal{B}$.)

(b) They tell us about the representation theory of G and $\mathfrak{g}$. For example, there is a bijection

$$\{\text{finite dim. } \mathbb{C}\text{-reps of } SL_n(\mathbb{C})\} \longleftrightarrow \{\text{finite dim. } \mathbb{C}\text{-reps of the Lie algebra of } SL_n(\mathbb{C})\}$$

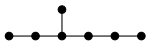
Where the Lie algebra of $SL_n(\mathbb{C})$ is

$$\mathfrak{sl}_n(\mathbb{C}) = \{A \in M_n(\mathbb{C}) |\ \mathrm{tr}(A) = 0\}.$$

We will be able to describe the RHS completely (by listing all the representations).

(c) They have applications in Algebraic Geometry. We can use Lie Algebras to build families of surfaces, or algebraic curves (See Slodowy's book for more).

We define the Dynkin diagram of a semisimple Lie algebra, e.g.



(This is the Dynkin diagram of "Type E_7"). This tells you about singularities on surfaces.

(d) There are applications to Number theory: root systems and Weyl groups can help you to get the structure of groups over $\mathbb{Q}_p$ (the p-adics).[1]

There is the Local Langlands correspondence which predicts a relationship:

$$\text{Galois Theory of Local Fields} \longleftrightarrow \text{Complex Lie Theory}$$

(e) There are also applications to applied maths, and that's all we'll say about that.

2 Basic Definitions and Examples

Let k be a field (most of the time this will be $\mathbb{C}$, but not always: we'll sometimes look at how things go wrong in fields with $\mathrm{char}\, k > 0$.)

Definition 2.1 (Lie Algebra)**.** A **Lie algebra** over k is a vector space $\mathfrak{g}$ over k with a bilinear pairing

$$[\,\cdot\,,\,\cdot\,] : \mathfrak{g} \times \mathfrak{g} \to \mathfrak{g}$$

which satisfies

- $[x, x] = 0$
- $[x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0$ (Jacobi Identity)

[1] See the Iwahari-Matsumoto Paper

Exercise: Check that these properties imply antisymmetry

$$[x, y] + [y, x] = 0$$

Definition 2.2 (Lie subalgebra)**.** A **Lie subalgebra** is a k-subspace $\mathfrak{h} \subset \mathfrak{g}$, such that $\mathfrak{h}$ is closed under the Lie bracket of $\mathfrak{g}$:

$$[x, y] \in \mathfrak{h} \qquad \forall x, y \in \mathfrak{h}$$

Example 2.3. Let V be finite-dimensional over k.

(a) Let $\mathfrak{gl}_n(V) = \mathrm{End}_k(V)$ with bracket

$$[x, y] = xy - yx.$$

Then it is clear that $[x, x] = 0$, and you can check Jacobi yourself.

If we choose a basis for V, then we can identify $\mathfrak{gl}_n$ with the space of $n \times n$ matrices over k, and then we can write $\mathfrak{gl}(V)$ as $\mathfrak{gl}_n$ (or $\mathfrak{gl}_n(k)$ to emphasise the field, or just $M_n(k)$).

(b) Let

$$\mathfrak{sl}(V) = \{x \in \mathfrak{gl}(V) \,:\, \mathrm{tr}(x) = 0\}.$$

This is clearly a subspace, since tr is linear. It is also closed under the Lie bracket, since tr is symmetric. That is to say:

$$\mathrm{tr}(xy - yx) = \mathrm{tr}(xy) - \mathrm{tr}(yx) = \mathrm{tr}(xy) - \mathrm{tr}(xy) = 0.$$

Then $\dim(\mathfrak{sl}(V)) = n^2 - 1$. And the standard basis for this is given by:

$$\{E_{ii} - E_{i+1,\,i+1} \,:\, 1 \leqslant i \leqslant n-1\} \cup \{E_{ij} \,:\, i \neq j\}.$$

Lecture 1

(c) Assume $\mathrm{char}\, k \neq 2$. Suppose that V is endowed with a symmetric bilinear form

$$[\,\cdot\,,\,\cdot\,] : V \times V \to k$$

Let

$$\mathfrak{so}(V) = \{X \in \mathfrak{gl}(V) \,:\, \langle Xv, w\rangle = -\langle v, Xw\rangle \,\forall v, w \in V\}$$

If we use coordinates, then we have $\exists M \in GL(V)$ such that $\langle v, w\rangle = v^{\mathrm{t}} M w$, so then we have

$$\mathfrak{so}(V) = \{X \,:\, MX + X^{\mathrm{T}} M = 0\}.$$

Usually, we take

$$M = \begin{cases} \begin{bmatrix} 0 & I_l \\ I_l & 0 \end{bmatrix}, & n = 2l \\ \\ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & I_l \\ 0 & I_l & 0 \end{bmatrix}, & n = 2l+1 \end{cases}$$

These are the *orthogonal Lie algebras*, denoted $\mathfrak{so}_n$.

Exercise: What is their dimension?

(A note: $\mathfrak{o}_n$ and $\mathfrak{so}_n$ are the same thing, since we "take tangent spaces at the identity," but the structure of $O(n)$ and $SO(n)$ do not differ at the identity, since the difference between them comes about because $O(n)$ has two connected components, so their Lie algebras are, in fact, identical).

Remark 2.4. If we let $n = 2$, and take

$$e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

viewed as matrices (in $\mathfrak{sl}_2(\mathbb{C})$). This gives the standard basis from example (2). Then we have

$$[e, f] = h,\ [h, e] = 2e,\ [h, f] = -2f,$$

and we will see that (in some sense) the structure of all semisimple Lie algebras 'comes from' $\mathfrak{sl}_2(\mathbb{C})$.

(d) Assume $\operatorname{char} k \neq 2$. Suppose V has a skew-symmetric non-degenerate bilinear form:

$$\langle \cdot, \cdot \rangle : V \times V \to k$$

i.e. such that

$$\langle v, w \rangle = -\langle w, v \rangle$$

Then

$$\mathfrak{sp} = \{x \in \mathfrak{gl}(V) \,:\, \langle Xv, w \rangle = \langle v, Xw \rangle\}$$

In coordinates, we take $\langle \cdot, \cdot \rangle$ to be the skew-symmetric form associated to:

$$\begin{pmatrix} 0 & I_l \\ -I_l & 0 \end{pmatrix}, \text{ where } n = 2l$$

This is the *symplectic Lie algebra*, denoted $\mathfrak{sp}_n$ (where $n = 2l$).

Note: n *must* be even, in order for V to have a skew-symmetric bilinear form.

Exercise: Check (3) and (4) are indeed Lie subalgebras of $\mathfrak{gl}(V)$.

(e) Any space V is a Lie algebra if we specify that

$$[v, w] = 0, \ \forall v, w \in V$$

These are called *abelian* Lie algebras. Named for obvious reasons, (i.e. for linear Lie algebras, it means that matrices commute).

(f) $\mathfrak{b}_n$: The set of upper triangular matrices (including the diagonal, i.e. non-strict upper triangular). This is the Borel subalgebra associated to the Borel subgroup of GL_n.

Exercise: What is the dimension of this Lie algebra, and why is it $\binom{n+1}{2}$?

(g) $\mathfrak{n}_n$ The set of *strictly* upper-triangular matrices. (The $\mathfrak{n}$ stands for nilpotent.)

3 Basic Representation Theory

Definition 3.1 (Lie Algebra Homomorphism)**.** A linear transformation

$$\phi : \mathfrak{g} \to \mathfrak{h}$$

between two Lie algebras is a **homomorphism** if:

$$[\phi(x), \phi(y)] = \phi([x, y]) \ \forall x, y \in \mathfrak{g}$$

If ϕ is also bijective (i.e. is an isomorphism of vector spaces) then it is a Lie algebra **isomorphism**.

Definition 3.2 (Representation)**.** Given a Lie algebra $\mathfrak{g}$, we say that a **representation** of $\mathfrak{g}$ is a Lie algebra homomorphism

$$\varphi : \mathfrak{g} \to \mathfrak{gl}(V)$$

for some V. (We can write $\varphi = \varphi_V$ to emphasise the space).

Notation:

- We also call V itself a 'representation,' or sometimes a $\mathfrak{g}$-module.
- We write: $G \curvearrowright V$ and say $\mathfrak{g}$ acts on V (on the left).
- We usually write $x.v$ or xv to mean $\varphi(x)(v)$
- The 'dimension' of the representation is $\dim V$.

Examples 3.3.

1. The trivial representation: Let $\dim V = 1$, and let $\mathfrak{g} \curvearrowright V$ via:

$$x.v = 0 \ \forall x \in \mathfrak{g}, v \in V$$

We call this representation the *trivial representation*

2. The defining representation: If $\mathfrak{g}$ is a subalgebra of $\mathfrak{gl}(V)$, then $\exists$ a natural inclusion:

$$\iota : \mathfrak{g} \to \mathfrak{gl}(V).$$

We call this inclusion the *defining representation* of $\mathfrak{g}$.

3. The adjoint representation: Let $x \in \mathfrak{g}$. Define

$$\begin{aligned} \operatorname{ad}(x) : \mathfrak{g} &\to \mathfrak{g}, \\ y &\mapsto [x, y], \end{aligned}$$

then the map

$$\begin{aligned} \mathfrak{g} &\to \mathfrak{gl}(\mathfrak{g}) \\ x &\mapsto \operatorname{ad} x \end{aligned}$$

is the adjoint map. (Exercise: Show this is a Lie algebra homomorphism – Use Jacobi, one line).

Note: Recall the earlier remark, the adjoint representation of $\mathfrak{sl}_2(\mathbb{C})$, with basis $\{e, h, f\}$ in that order, has:

$$\begin{aligned} \operatorname{ad} e &= \begin{bmatrix} 0 & -2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \\ \operatorname{ad} h &= \begin{bmatrix} 2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -2 \end{bmatrix} \\ \operatorname{ad} f &= \begin{bmatrix} 0 & 0 & 0 \\ -1 & 0 & 0 \\ 0 & 2 & 0 \end{bmatrix} \end{aligned}$$

4. If V, W are representations, then so is $V \oplus W$ via:

$$x.(v, w) = (xv, xw).$$

5. If V is a representation of $\mathfrak{g}$, then so is V^* via:

$$\forall x \in \mathfrak{g},\ v \in V,\ f \in V^*;\ (xf)(v) \to -f(x.v)$$

Check this.

6. If V, W are representations of $\mathfrak{g}$, then so is $\mathrm{Hom}(V, W)$ via:

$$\forall v \in V,\ x \in \mathfrak{g},\ f \in \mathrm{Hom}(V, W);\ (x.f)(v) = x.f(v) - f(x.v)$$

Check this too.

Definition 3.4 ($\mathfrak{g}$-equivariance)**.** If V, W are reps of $\mathfrak{g}$, then a linear transformation

$$\varphi : V \to W$$

is called **$\mathfrak{g}$-equivariant** if

$$\forall x \in \mathfrak{g},\ v \in V,\ x.\varphi(v) = \varphi(x.v).$$

We say V, W are isomorphic as representations if there is a $\mathfrak{g}$-equivariant isomorphism between them.

Definition 3.5 (Subrepresentation)**.** A **subrepresentation** $V' \subseteq V$ is a subspace such that

$$\forall x \in \mathfrak{g},\ v \in V';\ x.v \in V'.$$

Definition 3.6 (Irreducible)**.** We say that a representation is **irreducible** or **simple** if the only subrepresentations are V & $\mathbf{0}$ (but the $\mathbf{0}$ vector space is *not* an irreducible representation).

Exercise(s)

- The trivial representation is irreducible.
- The defining and adjoint representations of $\mathfrak{sl}_2(\mathbb{C})$ are irreducible.

Lecture 2

Definition 3.7 (Completely reducible/Semisimple)**.** V is a **completely irreducible** or **semisimple** representation if it decomposes as a sum of irreducible representations.

Exercise: For V a representation, complete reducibility is equivalent to

$$\forall \text{subreps } W \subseteq V,\ \exists W' \text{ such that } V' = W \oplus W'.$$

If V is a representation, $W \leqslant V$ a subrepresentation, then

$$\frac{V}{W}$$

is a representation of $\mathfrak{g}$ via

$$x : (v + W) = xv + W$$

4 Representation Theory of $\mathfrak{sl}_2(\mathbb{C})$

Recall that

$$\mathfrak{sl}_2(\mathbb{C}) = \{X \,:\, \mathrm{tr}(X) = 0\}.$$

Example: Consider the Borel subalgebra of dimension 2, given by

$$\mathfrak{b}_2 = \{\begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \,:\, a, b, c \in \mathbb{C}\}.$$

Then let $V = \mathrm{span}\{\begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix}\}$, the defining representation of $\mathfrak{b}_2$. This is not completely irreducible (there is a similar example in II Representation theory). $\left\langle \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\rangle$ is a subrepresentation, but there is no complement subrepresentation by easy linear algebra.

Definition 4.1 (Faithful)**.** A representation V of a Lie algebra $\mathfrak{g}$ is **faithful** if

$$\mathfrak{g} \to \mathfrak{gl}(V)$$

is injective.

Henceforth, all Lie algebras and their representations are over $\mathbb{C}$ (unless stated otherwise).

Let V be a representation of $\mathfrak{sl}_2(\mathbb{C})$, recall:

$$e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

We know three representations of $\mathfrak{sl}_2(\mathbb{C})$ already:

Dim	Name	Action of h
1	trivial	0
2	defining	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$
3	adjoint	$\begin{pmatrix} 2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -2 \end{pmatrix}$

Definition 4.2 (λ-weight space)**.** For $\lambda \in \mathbb{C}$ the **λ-weight space** of V is

$$V_\lambda = \{v \in V \,:\, hv = \lambda v\}.$$

The following are *vector space sums* of *weight-spaces*, **not** decompositions into irreducible representations.

- The trivial is V_0
- The defining is $V_1 \oplus V_{-1}$
- The adjoint is $V_2 \oplus V_0 \oplus V_{-2}$. Given by $\langle e \rangle \oplus \langle h \rangle \oplus \langle f \rangle$

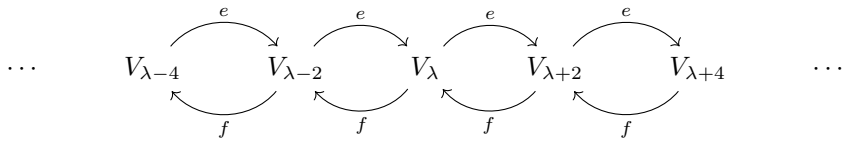
Actions:

Of e: for $v \in V_\lambda$,

$$\begin{aligned} h \cdot (ev) &= ([h, e] + eh)v \\ &= (2ev) + \lambda ev \end{aligned}$$

So $h \cdot ev = (\lambda + 2)ev$. So $ev \in V_{\lambda+2}$.

Of f: By a virtually identical argument, $fv \in V_{\lambda-2}$. So we have the following diagram:



Definition 4.3 (Highest Weight Vector)**.** If a non-zero $v \in V_\lambda \cap \ker(e)$ for some λ, then v is called a **Highest Weight Vector** (or HWV) of weight λ. That is to say:

$$hv = \lambda v \qquad\qquad ev = 0$$

Example: In the adjoint representation, e is a highest weight vector of weight 2.

Lemma 4.4. *Suppose $v \in V_\lambda$ is a HWV, then $\forall n \geqslant 1$*

$$ef^n v = n(\lambda - n + 1)f^{n-1}v$$

Proof. Easy induction on n

Base Case:

$$(ef)v = ([e, f] + fe)v = (h + fe)v = hv = \lambda v.$$

Inductive Step: Suppose this is true for $n = k - 1$, then we have:

$$ef^{k-1}v = (k-1)(\lambda - (k-1) + 1)f^{k-2}v.$$

Then:

$$\begin{aligned} ef^k v &= ([e,f]f^{k-1} + fef^{k-1})v \\ &= (h + fe).(f^{k-1}v) \\ &= (hf^{k-1}v) + (k-1)(\lambda - (k-1) + 1)f^{k-1}v \\ &= (\lambda - 2(k-1))f^{k-1}v + (k-1)(\lambda - k + 2)f^{k-1}v \\ &= k(\lambda - k + 1)f^{k-1}v \end{aligned}$$

□

Lemma 4.5. *Suppose $v \in V_\lambda$ is a HWV. Then $W = \text{span}\{v, fv, f^2v, \dots\}$ is a subrepresentation of V.*

Proof. It suffices to show that if $w = f^n v$, then

(i) $ew \in W$

(ii) $hw \in W$

(iii) $fw \in W$

(iii) Clearly true by definition of W.

(i) If $n > 1$ this follows from Lemma 4.4 above, and if $n = 0$ then $ew = 0$ by definition of HWV.

(ii) $f^n w \in V_{\lambda - 2n}$, so $hw = (\lambda - 2n)w \in W$. □

Proposition 4.6. *If V is finite dimensional, it contains at least one highest weight vector (no claim about uniqueness).*

Proof. Choose any non-zero eigenvector v of h. (This can always be done, since we're in $\mathbb{C}$ now).

Consider $v, ev, e^2v, \dots$. This set:

$$\{e^n v \,:\, ev \neq 0\}$$

is linearly independent, as the e action changes the eigenvalue of v when acted on by h. Therefore. as V is finite dimensional, this set must be finite. So $\exists\, n$ such that $e^n v \neq 0$, but $e^{n+1}v = 0$. Therefore $e^n v$ is a highest weight vector, with weight $\lambda + 2n$.

Lemma 4.7. *Suppose V is finite dimensional, and $v \in V_\lambda$ is a highest weight vector, then $\lambda \in \mathbb{Z}_{\geqslant 0}$.*

Proof. Any non-zero vectors of the form $f^n v$ must be linearly independent. So

$$\exists n \geqslant 0 \text{ s.t } f^n v \neq 0 \text{ but } f^{n+1} v = 0$$

By (4.4)

$$0 = (ef^{n+1})v = (n+1)(\lambda - n) f^n v$$

Therefore $f^n v \neq 0$, $n+1 \neq 0$, so $\lambda = n$. □

Suppose that V is irreducible, of dimension $n+1$. Then by (4.6), there is a HWV $v \in V_\lambda$, by (4.5), the span of $\{v, fv, f^2, \dots\}$ is a subrepresentation of V. So $\{v, fv, f^2 v, \dots, f^n v\}$ is a basis, as the $f^i v$ are all linearly independent for $0 \leqslant i \leqslant n$. We know from (4.7) that $\lambda = n \in \mathbb{Z}_{\geqslant 0}$.

Corollary 4.8. *If V an irreducible representation of $\mathfrak{sl}_2(\mathbb{C})$ of dimension $n+1$, then there exists a basis $v_0, v_1, \dots v_n$ of v such that*

(i) $hv_i = (n-2i)v_i$, that is:

$$h = \begin{bmatrix} n & & & & \\ & n-2 & & & \\ & & \ddots & & \\ & & & 2-n & \\ & & & & n \end{bmatrix}$$

(ii) $f_{v_i} = v_{i+1}$ for $0 \leqslant i \leqslant n-1$, that is:

$$f = \begin{bmatrix} 0 & & & & \\ 1 & 0 & & & \\ & 1 & 0 & & \\ & & \ddots & \ddots & \\ & & & 1 & 0 \end{bmatrix}$$

(iii) $ev_i = \begin{cases} i(n-i+1)v_{i-1}, & i \geqslant 1 \\ 0, & i = 0 \end{cases}$, *that is:*

$$e = \begin{bmatrix} 0 & n & & & \\ & 0 & 2(n-1) & & \\ & & 0 & \ddots & \\ & & & \ddots & n \\ & & & & 0 \end{bmatrix}.$$

Lecture 3

5 Irreducible modules for $\mathfrak{sl}_2$

Remark 5.1. Let V be an (n+1)-dimensional irreducible representation of $\mathfrak{sl}_2$, $v \in V$ a highest weight vector. Then note:

$$(ef + fe + \frac{1}{2}h^2)(v) = nv + \frac{n^2}{2}v = (\frac{n^2}{2} + n)v.$$

This is a special operator, and we're going to see it three times very quickly.

We write $V(n)$ for the unique irreducible representation of $\mathfrak{sl}_2$ of dimension $n+1$.

Definition 5.2 (Weights)**.** Given a representation V of $\mathfrak{sl}_2$, then

$$\{\lambda \in \mathbb{C} : V_\lambda \neq 0\}$$

are called the **weights** of V.

Today, we will show Weyl's theorem which states that the category of finite dimensional representations of $\mathfrak{sl}_2(\mathbb{C})$ is irreducible.

Theorem 5.3. *Every finite dimensional representation of $\mathfrak{sl}_2$ is completely reducible.*

This result, along with (4.8), says that to understand a representation of $\mathfrak{sl}_2$, you essentially just need to understand the action of h. That is, the action of h completely determines a finite dimensional representation of $\mathfrak{sl}_2(\mathbb{C})$.

Example 5.4. Suppose you have some 5-dimensional representation of $\mathfrak{sl}_2$, and that $\exists v \in V$ such that $hv = 3v$. This means that the weights will contain $3, 1, -1, -3$, so we must have

$$V = V(0) \oplus V(3),$$

since $5 = 1 + 4$. (Dimension counting.) □

Before we prove (5.2), we need some facts. First of all, if $\mathfrak{g}$ is an arbitrary Lie algebra, and

$$\varphi : \mathfrak{g} \to \mathfrak{gl}(V)$$

is a representation of $\mathfrak{g}$, and there is some σ which commutes with $\varphi(x)$ $\forall x \in \mathfrak{g}$. Then we have facts, which you should check:

1. $\ker(\sigma - \lambda \operatorname{Id}_V)$ is a subrepresentation of V for all $\lambda \in \mathbb{C}$.

2. If V is irreducible, then σ is a scalar multiple of the identity. (Schur's Lemma).

Definition 5.5 (Casimir element)**.** Let V be a finite dimensional representation of $\mathfrak{sl}_2$. Then

$$\Omega = (ef + fe + \frac{1}{2}h^2) \in \mathfrak{gl}(V)$$

is called the **Casimir element** (of $\mathfrak{sl}_2$). In fact, Ω is central.

Lemma 5.6 (The Casimir element is central)**.** *If*

$$\varphi : \mathfrak{sl}_2 \to \mathfrak{gl}(V),$$

then Ω commutes with $\varphi(x)$, $\forall x \in \mathfrak{sl}_2$ (for V finite dimensional).

Proof. To show Ω is central, show $e\Omega = \Omega e$, $f\Omega = \Omega f$, $h\Omega = \Omega h$. Just a mechanical check. □

Corollary 5.7. *If V is an irreducible finite dimensional representation of $\mathfrak{sl}_2$ then Ω acts on V by a scalar.*

Proof. Schur's Lemma. By the remark above, the scalar is $(\frac{n^2}{2} + n)$. □

Proof of (5.2). The general case is done in Humphreys' book on page 63. There's a different proof in Henderson's book: 5.2.1, 7.5.1.

Let

$$\varphi : \mathfrak{sl}_2 \to \mathfrak{gl}(V)$$

be a finite dimensional representation of $\mathfrak{sl}_2$. Let $W \leqslant V$ be a subrepresentation. Then we need to show $\exists U \leqslant V$ such that

$$V = W \oplus U.$$

Case 1. W has codimension 1[2]. So we must have $\frac{V}{W} \cong V(0)$.

Subcases:

[2] $\frac{V}{W}$ has dimension 1

(1) W is trivial, so $\dim V = 2$, and there exists a basis of V (v_1, v_2 such that $W = \mathbb{C}v_1$) with respect to which $\mathfrak{sl}_2$ acts on V by matrices which look like:

$$\begin{pmatrix} 0 & * \\ 0 & 0 \end{pmatrix}.$$

We show $V \cong V(0) \oplus V(0)$ where the first $V(0)$ is W and the second one is the U that we are looking for.
Note:

$$\left[\begin{pmatrix} 0 & x \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & y \\ 0 & 0 \end{pmatrix}\right] = 0 \ \forall x, y \in \mathbb{C}$$

Since φ respects the bracket:

$$\varphi(h) = [\varphi(e), \varphi(f)] = 0,$$

$$\varphi(e) = \frac{1}{2}[\varphi(h), \varphi(e)] = 0,$$

$$\varphi(f) = \frac{1}{2}[\varphi(h), \varphi(f)] = 0.$$

So in other words, the action *must* be trivial on the other factor, and that is the complement that we were after.

(2) $W \cong V(n)$ is irreducible, but nontrivial ($n > 0$). Now we have:

$$\Omega \in \mathfrak{gl}(V)$$

We show $V = V(n) \oplus \ker(\Omega)$.

By Schur, and the fact that W is irreducible, *and* the fact that Ω acts on the quotient trivially (since codimension is 1), there is a basis for V such that Ω acts via:

$$\begin{pmatrix} \lambda & & & & * \\ & \lambda & & & * \\ & & \ddots & & * \\ & & & \lambda & * \\ & & & & 0 \end{pmatrix}$$

W is nontrivial, so $\ker \Omega \neq 0$, and it's now clear that

$$W \cap \ker \Omega = 0.$$

So in other words,

$$V = W \oplus \ker \Omega.$$

(3) Now let's take an arbitrary W. The idea now is to do induction on the $\dim V$. If $\dim V = 1$ then it is trivial, so that is already taken care of in subcase 1. So now assume $\dim V \geqslant 2$.

Take some subrepresentation $0 \neq W' \leqslant W$. As $\dim\left(\frac{W}{W'}\right) < \dim V$, and the codimension of $\frac{W}{W'}$ in $\frac{V}{W'}$ is 1 (by, like, the third(?) isomorphism theorem, maybe it's the second), then by induction, we have:

$$\frac{V}{W'} = \frac{W}{W'} \oplus \frac{W''}{W'} \tag{1}$$

for some subspace W'' of V which contains W', and is such that $\frac{W''}{W'}$ is a subrepresentation of $\frac{V}{W'}$.

Then $W' \leqslant W''$ has codimension 1 and morover $\dim W' < \dim V$. W'' is a subrepresentation of V, since $\frac{W''}{W'}$ being a subrepresentation implies that

$$xw \in W' \leqslant W'' \ \forall x \in \mathfrak{sl}_2.$$

Again by induction, $\exists U \leqslant W''$ such that

$$W'' = W' \oplus U. \tag{2}$$

Then we claim that

$$V = W \oplus U.$$

We know already that $\dim U = 1$, and that $W \cap U \leqslant W \cap W'' = W'$, since we showed that (1) is a direct sum. But, by (2), as this sum is direct, $W \cap U \leqslant W' \cap U = 0$, and since $\dim U = 1$ and the codimension of W in V is 1, we're done by counting dimensions (rank-nullity).

Case 2. Let W be arbitrary. Recall from chapter 3 the action on $\mathrm{Hom}(V, W)$:

$$(x\varphi)(v) = x(\varphi(v)) - \varphi(xv) \tag{3.3(6)}$$

Now we define

$$\begin{aligned}\mathbf{V} &= \{\psi \in \mathrm{Hom}(V, W) \,:\, \psi|_W = \lambda \,\mathrm{Id}_W \text{ for some } \lambda \in \mathbb{C}\},\\ \mathbf{W} &= \{\psi \in \mathbf{V} \,:\, \psi|_W = 0\}.\end{aligned}$$

Note that the codimension of $\mathbf{W}$ in $\mathbf{V}$ is 1, and suppose

$$\psi|_W = \lambda \,\mathrm{Id}_W,\ x \in \mathfrak{sl}_2,\ w \in W.$$

Then

$$\begin{aligned}(x\psi)(w) = x\psi(w) - \psi(xw) &= x(\lambda w) - \lambda(xw)\\ &= 0.\end{aligned}$$

So $\mathbf{V}$ is a subrepresentation of $\mathrm{Hom}(V, W)$ threfore $\mathbf{W}$ is a subrepresentation. By case 1, $\exists 1-$dim subrepresentation $\mathbf{U}$ of $\mathbf{V}$ such that

$$\mathbf{V} = \mathbf{W} \oplus \mathbf{U}.$$

Write $\mathbf{U} = \langle \gamma \rangle$ for some $\gamma \in \mathbf{V}$, so that $\gamma|_W = \lambda \,\mathrm{Id}_W$, for some $0 \neq \lambda \in \mathbb{C}$. The claim is that (as vector spaces, and then as representations):

$$V = W \oplus \ker(\gamma).$$

Lecture 4 where, we claim, that $U = \ker \gamma$. $W \cap \ker \gamma = 0$, so by dimension counting,

$$\dim V = \dim W + \dim \ker \gamma.$$

as $W = \mathrm{Im}\, \gamma$. So we *do* have a direct sum of vector spaces. Finally, we show that $\ker \gamma$ is a subrepresentation of V. Let $v \in \ker \gamma$ and take $x \in \mathfrak{sl}_2$. $\mathbf{U}$ is a 1-dimensional representation of $\mathfrak{sl}_2$, so it must be the trivial representation, since this is the only 1-dim representation of $\mathfrak{sl}_2$. Then we have

$$\begin{aligned} 0 = (x\gamma)(v) &= x\gamma(v) - \gamma(xv) \\ \Rightarrow x\gamma(v) &= \gamma(xv) \\ \Rightarrow 0 &= \gamma(xv) \qquad (v \in \ker \gamma) \end{aligned}$$

as required, since then $xv \in \ker \gamma$ for any $x \in \mathfrak{sl}_2$, so it must be a subrepresentation.

Remark 5.8. Proof only needed:

- The existence of Ω, the Casimir element.
- Every 1-dimensional representation of $\mathfrak{sl}_2$ is the trivial representation.

Remark 5.9. Complete reducibility is *rare*. It can fail for infinite dimensional Lie algebras, or *simple Lie algebras* in positive characteristic fields.

Exercise: Show that the adjoint representation of $\mathfrak{sl}_n(\mathbb{F}_p)$ on $gl_n(\mathbb{F}_p)$ is not completely reducible, provided $p|n$. (Quite a hard exericse, apparently).

6 Tensor products

Given two finite dimensional vector spaces V, W; with bases $\{v_1, \ldots, v_k\}$, and $\{w_1, \ldots, w_m\}$ respectively. Recall that

$$V \otimes_{\mathbb{C}} W$$

has basis

$$\{v_i \otimes w_j \,:\, 1 \leqslant i \leqslant k,\, 1 \leqslant j \leqslant m\}.$$

Definition 6.1 (Tensor representation)**.** If V, W are representations of a Lie algebra $\mathfrak{g}$, then so is $V \otimes W$ with

$$x \cdot (v \otimes w) = (x \cdot v) \otimes w + v \otimes (x \cdot w).$$

Examples 6.2.

(1) If V, W representations of $\mathfrak{sl}_2$, $v \in V_\lambda$, $w \in W_\mu$, then

$$h \cdot (v \otimes w) = (\lambda + \mu)(v \otimes w).$$

In particular, the weights of $V \otimes W$ are:

$$\{\lambda + \mu \,:\, \lambda \text{ a weight for } V, \mu \text{ a weight for } W\}.$$

(2) Decomposition of $V(2) \otimes V(2) = V(2)^{\otimes 2}$:

	2	0	-2
2	4	2	0
0	2	0	-2
-2	0	-2	-4

So $V(2)^{\otimes 2} = V(4) \oplus V(2) \oplus V(0)$.

(3) If v_n is a highest weight vector in $V(n)$, show that $v_n \otimes v_m$ is the highest weight vector in $V(n) \otimes V(m)$.

Question: Is there a general formula for decomposing $V(n) \otimes V(m)$?

Answer: Clebsch-Gordon formula

$$V(n) \otimes V(m) = \bigoplus_{r=|n-m|}^{n+m} V(r),$$

going up in 2s[3].

Definition 6.3 (Symmetric power)**.** The **nth Symmetric power is**

$$S^n(V) = \mathrm{Sym}^n(V) = SV = \frac{V \otimes V \otimes \cdots \otimes V}{M_n}$$

where $M_n = \langle u_1 \otimes \cdots \otimes u_n - u_{\sigma(1)} \otimes \cdots \otimes u_{\sigma(n)} \,|\, \sigma \in S_n, u_i \in V \rangle$.

For example $n = 2$, have $M_2 = \{v \otimes u - u \otimes v\}$.

[3]See Part II Representation Theory for more information

Fact. M_n is a subrepresentation of $V \otimes \cdots \otimes V$ whenever V is a subrepresentation of $\mathfrak{g}$, so $\mathrm{Sym}^n(V)$ is a subrepresentation. Prove this as an exercise.

Example 6.4. In S^2V, $v \otimes w = w \otimes v$. Therefore S^2V has basis

$$\{v_i \otimes v_j \,:\, i \leqslant j\}.$$

Decomposing $S^2(V(2))$, we have $0 \neq e \otimes e \in S^2(V(2))$, so $V(4)$ is a subrepresentation, and in fact

$$S^2(V(2)) = V(4) \oplus V(0)$$

Definition 6.5 (Exterior/Alternating Power)**.** The **nth exterior/alternating power** is

$$\Lambda^n V = \frac{V^{\otimes n}}{N_n}$$

where $N_n = \langle u_1 \otimes \cdots \otimes u_n \,:\, u_i \in V \,\forall i,\, u_i = u_j \text{ for some j} \rangle$.

Example 6.6. With $n = 2$, we have $N_2 = \langle v \otimes v \,:\, v \in V \rangle$. N_n is a subrepresentation of $V^{\otimes n}$. Prove this.

Notation. Write (the coset of) $u_1 \otimes \cdots \otimes u_n$ as $u_1 \wedge \cdots \wedge u_n$.

Decompose

$$\Lambda^2(V(2)) \cong V(2).$$

with basis

$$\{e \wedge f, e \wedge h, h \wedge f\}.$$

Exercise. Find the dimensions of symmetric/exterior powers.

7 Results about semisimple Lie algebras

First, let $\mathfrak{g}$ be a semisimple Lie algebra over $\mathbb{C}$.

Definition 7.1 (Ideal)**.** A subspace $I \subseteq \mathfrak{g}$ is an **ideal** of $\mathfrak{g}$ if

$$[x, y] \in I,\ \forall x \in \mathfrak{g},\, y \in I.$$

- Any ideal is a subalgebra
- If I is an ideal, then $\mathfrak{g}/I$ is a Lie algebra with bracket:

$$[x + I, y + I] = [x, y] + I.$$

- I is an ideal if and only if it is a subrepresentation of the adjoint representation of $\mathfrak{g}$.

Examples.

(1) The centre of $\mathfrak{g}$ is

$$Z(\mathfrak{g}) = Z = \{x \in \mathfrak{g} \,:\, [xy] = 0, \forall y \in \mathfrak{g}\}.$$

Check that this is an ideal (use Jacobi identity). Note $Z = \ker(\mathrm{ad} : \mathfrak{g} \to \mathrm{End}\,\mathfrak{g})$.

Remark. If $Z(G) = 0$, then the adjoint is faithful, and $\mathfrak{g}$ can be embedded inside $\mathrm{End}\,\mathfrak{g}$, so it can be regarded as a Lie subalgebra of $\mathfrak{gl}_n$ for some n. [4]

(2) The derived subalgebra is

$$D(\mathfrak{g}) = \langle [x, y] \,|\, x, y \in \mathfrak{g} \rangle.$$

This is an ideal. Recall that $\mathfrak{sl}_n$ is a subalgebra of $\mathfrak{gl}_n$. Show that, in fact, $\mathfrak{sl}_n$ is the *derived subalgebra* of $\mathfrak{gl}_n$.

(3) If

$$\phi : \mathfrak{g} \to \mathfrak{h}$$

is a homomorphism. Then $\ker \phi$ is also an ideal (this shouldn't be too surprising).

Lecture 5

Definition 7.2 (Simple Lie algebra)**.** The Lie algebra $\mathfrak{g}$ is **simple** if

$$[\mathfrak{g}, \mathfrak{g}] \neq 0$$

and the only ideals of $\mathfrak{g}$ are 0 and $\mathfrak{g}$.

Examples.

- We can show $\mathfrak{sl}_n$ is simple for $n \geqslant 2$.
- $\mathfrak{so}_n$ is simple for $n \geqslant 5$.
- $\mathfrak{sp}_{2l}$ for $l \geqslant 1$ is simple.

Try this now for $\mathfrak{sl}_2(\mathbb{C})$.

Note.

- $\mathfrak{g}$ simple $\Rightarrow [\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}$.

[4] Therefore is a deep theorem of Ado (not proved here) that says that if $\mathrm{char}\, k = 0$ then any finite dimensional Lie algebra $\mathfrak{g}$ embeds as a Lie subalgebra in $\mathfrak{gl}_n$ for some m, but this embedding is necessarily from the adjoint. In fact, this is true in characteristic p as well, due to Iwasawa.

- $\mathfrak{g}$ simple $\Rightarrow$ every representation of $\mathfrak{g}$ is either faithful or a direct sum of trivial representations.

So $\mathfrak{g}$ simple $\Leftrightarrow$ the adjoint representation is irreducible.

Definition 7.3 (Semisimple Lie algebra)**.** A Lie algebra $\mathfrak{g}$ is **semisimple** if it is the direct sum of simple ideals. I.e. ideals which are themselves simple when viewed as Lie algebras.

For example $\mathfrak{so}_4 \cong \mathfrak{sl}_2 \oplus \mathfrak{sl}_2$. We state a more "standard" definition of semisimple Lie algebras and then show it is equivalent to (7.3).

Definition 7.4 (Central series)**.** The **Central series** (or lower central series) is the sequence of subalgebras

$$\mathfrak{g}^0 \supseteq \mathfrak{g}^1 \supseteq \mathfrak{g}^2 \supseteq \dots,$$

where $\mathfrak{g}^0 = \mathfrak{g}$ and $\mathfrak{g}^n = [\mathfrak{g}, \mathfrak{g}^{n-1}]$. That is to say:

$$\mathfrak{g} \supseteq [\mathfrak{g}, \mathfrak{g}] \supseteq [\mathfrak{g}, [\mathfrak{g}, \mathfrak{g}]] \supseteq \dots.$$

Definition 7.5 (Derived series)**.** The **Derived series** (or upper central series) for $\mathfrak{g}$ is the sequence

$$\mathfrak{g}^{(0)} \supseteq \mathfrak{g}^{(1)} \supseteq \mathfrak{g}^{(2)} \supseteq \dots,$$

where $\mathfrak{g}^{(0)} = \mathfrak{g}$, $\mathfrak{g}^{(n)} = \mathfrak{g}^{(n-1)}, \mathfrak{g}^{(n-1)}$. That is to say

$$\mathfrak{g} \supseteq [\mathfrak{g}, \mathfrak{g}] \supseteq [[\mathfrak{g}, \mathfrak{g}], [\mathfrak{g}, \mathfrak{g}]] \supseteq \dots.$$

Note that:

- $\mathfrak{g}^{(n)} \subseteq \mathfrak{g}^n$
- $\forall n$ $\mathfrak{g}^{(n)}, \mathfrak{g}^n$ are ideals, we will prove this:

Proof. Induction on n, with $n = 0$ clear. Let x,y in $\mathfrak{g}$, $\mathfrak{g}^n$ respectively, then

$$[x, y] \in \mathfrak{g}^{n-1}.$$

Since $\mathfrak{g}^{n-1}$ is an ideal, and $\mathfrak{g}^n \subseteq \mathfrak{g}^{n-1}$. But

$$\begin{aligned} \mathfrak{g}^n &= \{[x, y] \,:\, x \in \mathfrak{g}, y \in \mathfrak{g}^{n-1}\} \\ &\supseteq \{[x, y] \,:\, x \in \mathfrak{g}, y \in \mathfrak{g}^n \end{aligned}$$

Similarly, do $\mathfrak{g}^{(n)}$.

Examples.

- If $\mathfrak{g}$ is simple, both series look like

$$\mathfrak{g} \supseteq \mathfrak{g} \supseteq \mathfrak{g} \supseteq \dots.$$

- If $\mathfrak{g}$ is abelian, both series look like

$$\mathfrak{g} \supseteq 0 \supseteq 0 \supseteq \dots.$$

- Let

$$\mathfrak{n}_n = \begin{pmatrix} 0 & * & * \\ & \ddots & * \\ 0 & 0 & 0 \end{pmatrix} \subseteq \mathfrak{gl}_n$$

  Central series for $\mathfrak{n}_n$:

$$\left\{ \begin{pmatrix} 0 & * & * & * \\ & \ddots & * & * \\ & & \ddots & * \\ 0 & 0 & 0 & 0 \end{pmatrix} \right\} \supseteq \left\{ \begin{pmatrix} 0 & 0 & * & * \\ & \ddots & \ddots & * \\ & & \ddots & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \right\} \supseteq \left\{ \begin{pmatrix} 0 & 0 & 0 & * \\ & \ddots & \ddots & 0 \\ & & \ddots & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \right\} \supseteq \dots$$

  If $n = 3$ this is the Heisenberg Lie algebra. Moreover, this central series eventually becomes 0, so we call it nilpotent, as a result of the following definition.

Definition 7.6 (Nilpotent/Solvable)**.** If $\mathfrak{g}^n = 0$ for some n, then $\mathfrak{g}$ is called **nilpotent**. If $\mathfrak{g}^{(n)} = 0$ for some n, then $\mathfrak{g}$ is called **solvable**.

Exercise. Show $\mathfrak{b}_n$ is solvable but not nilpotent ($n \geqslant 2$). By the note to (7.5), we have that nilpotent $\Rightarrow$ solvable.

Next result allows the theory of compex semisimple Lie algebras to go very far with minimal work.

Theorem 7.7 (Lie's theorem)**.** *Let $k = \mathbb{C}$. Let $\mathfrak{g} \leqslant \mathfrak{gl}(V)$ be a Lie subalgebra. Suppose that $\mathfrak{g}$ is solvable. Then there exists a common eigenvector for all the elements of $\mathfrak{g}$:*

$$\exists 0 \neq v \in V \ \ s.t. \ \ \forall x \in \mathfrak{g}, \ \exists \lambda_n \in \mathbb{C} \ \ s.t. \ \ xv = \lambda_n v$$

Proof. See Humphrey, Theorem 4.1. Just a two page linear algebra hack.

Corollary 7.8. *There exists a basis for V such that every element is upper triangular. In fact, using Lie's theorem and an inductive argument, we can show that there is a chain of subspaces of V such that*

$$0 = V_0 \leqslant V_1 \leqslant V_2 \leqslant \ldots \leqslant V_n = V,$$

with $\dim V_i = i$ and $\mathfrak{g}(V_i) \subseteq V_i$ for each i (this is called a maximal flag). Taking basis of V such that $V_i = \operatorname{span}(e_1, \ldots, e_i)$, we have that $\mathfrak{g}$ can be represented by upper triangular matrices, and so using such a basis

$$\mathfrak{g} \leqslant \mathfrak{b}_n.$$

Proof. Non-examinable. Probably also in Humphreys' book somewhere.

Proposition 7.9. *Suppose I, J ideals of $\mathfrak{g}$, then*

1. *If $\mathfrak{g}$ solvable, then any subalgebra, or quotient of $\mathfrak{g}$ is solvable.*
2. *If I is solvable, and $\frac{\mathfrak{g}}{I}$ is solvable, then so is $\mathfrak{g}$.*
3. *If I is solvable and J is solvable, then $I + J$ is solvable too.*

Proof.

1. Obvious from the definition of solvability.
2. Choose n such that
$$\left(\frac{\mathfrak{g}}{I}\right)^{(n)} = 0$$
So $\mathfrak{g}^{(n)} \subseteq I$. Now we have
$$\mathfrak{g}^{(n+m)} \subseteq I^{(m)}$$
for each $m \geqslant 0$, and since I is solvable, we're done.
3. Note that
$$\frac{(I+J)}{J} \cong \frac{I}{I \cap J}.$$
The RHS is solvable by 1, and J is solvable by assumption, and $\frac{(I+J)}{J}$ is therefore solvable. By 2, $I + J$ also solvable

□

Definition 7.10 (Radical)**.** The **Radical** of $\mathfrak{g}$, $\mathrm{Rad}(\mathfrak{g})$ is the maximal solvable ideal of $\mathfrak{g}$. It is the sum of all the solvable ideals of $\mathfrak{g}$.

Definition 7.11 (Trace form)**.** If $\phi : \mathfrak{g} \to \mathfrak{gl}(V)$ is a finite dimensional representation of $\mathfrak{g}$, then the **Trace form** of V:

$$\begin{aligned}(\,\cdot\,,\,\cdot\,) : \mathfrak{g} \times \mathfrak{g} &\to \mathbb{C} \\ (x, y) &\mapsto \mathrm{tr}(\phi(x)\phi(y))\end{aligned}$$

Exercise.

1. ϕ a representation of $\mathfrak{g} \Rightarrow (\,\cdot\,,\,\cdot\,)_V$ is symmetric and bilinear.
2. $([x, y], z)_V = (x, [y, z])_V$, called "Invariance."

Definition 7.12 (Killing form)**.** The **Killing form** $K(\,\cdot\,,\,\cdot\,)$ is the trace form attached to the adjoint representation,

$$K(x, y) = \mathrm{tr}(\mathrm{ad}(x), \mathrm{ad}(y)) = (x, y)_{\mathrm{ad}}.$$

Also, we have that

$$K([x, y], z) = K(x, [y, z]),$$

and we will use this a lot. (Easy to prove using linearity and cycle-invariance of trace).

Lecture 6

Theorem 7.13. *The following are equivalent for a finite dimensional Lie algebra* $\mathfrak{g}$

1. $\mathfrak{g}$ *is semisimple (as defined in 7.3)*
2. $\mathrm{Rad}(\mathfrak{g}) = 0$
3. *The Killing form on* $\mathfrak{g}$ *is non-degenerate. (Cartan-Killing criterion).*

Remark. $\mathrm{Rad}(\frac{\mathfrak{g}}{\mathrm{Rad}(\mathfrak{g})}) = 0$ (A solvable ideal of $\frac{\mathfrak{g}}{\mathrm{Rad}(\mathfrak{g})}$)) would lift to give an ideal J of $\mathfrak{g}$ containing $\mathrm{Rad}(\mathfrak{g})$ with $\frac{J}{\mathrm{Rad}(\mathfrak{g})}$ solvable

$$\begin{aligned}&\Rightarrow J \text{ solvable } \therefore J \subseteq \mathrm{Rad}(\mathfrak{g}) \\ &\Rightarrow \frac{\mathfrak{g}}{\mathrm{Rad}(\mathfrak{g})} \text{ semisimple}\end{aligned}$$

Levi's Theorem. (not proved here) Says that if $\operatorname{char} k = 0$ and $\mathfrak{g}$ is finite dimensional, then there is a Lie subalgebra $\mathfrak{g}'$ such that

$$\mathfrak{g}' \cap \operatorname{Rad}(\mathfrak{g}) = 0$$

and also

$$\mathfrak{g} = \mathfrak{g}' \oplus \operatorname{Rad}(\mathfrak{g}).$$

Thus

$$\mathfrak{g} \cong \frac{\mathfrak{g}}{\operatorname{Rad} \mathfrak{g}}$$

is a semisimple Lie algebra. What this is saying is that, considering the following short exact sequence.

$$0 \to \operatorname{Rad}(\mathfrak{g}) \to \mathfrak{g} \to \frac{\mathfrak{g}}{\operatorname{Rad}(\mathfrak{g})} \to 0$$

splits, i.e. there exists $\mathfrak{g}' \subseteq \mathfrak{g}$ which maps into $\frac{\mathfrak{g}}{\operatorname{Rad}(\mathfrak{g})}$. This is called the Levi decomposition and $\mathfrak{g}'$ is the Levi subalgebra of $\mathfrak{g}$.

This result fails in $\operatorname{char} k = p$ and for $\dim \mathfrak{g} = \infty$.

Lemma 7.14. *Let $\mathfrak{g}$ be a Lie algebra, then*

1. *If I is an ideal of $\mathfrak{g}$, then so is the derived algebra $[I, I]$*
2. *$\operatorname{Rad}(\mathfrak{g}) = 0 \Leftrightarrow \mathfrak{g}$ has no non-trivial abelian ideals.*

Proof.

1. If $x, y \in I$, $z \in \mathfrak{g}$, we need to show

   $$[z, [x, y]] \in [I, I].$$

   The obvious thing to try is Jacobi, which gets us

   $$[z, [x, y]] = -[x, [y, z]] - [y, [z, x]]$$

   then both summands on the right hand side have the inner bracket containing something in I, so the overall inner bracket is in I, and since these are bracketed with something which is also contained in I. Therefore, each overall expression is in $[I, I]$ Therefore the left hand side must also be in $[I, I]$.

2. It's clear that any abelian ideal must be solvable. For the other direction, we have that if I is solvable then the last non-zero term in the derived series for I will be abelian. (Done by contrapositive both ways).

□

Let

$$\mathfrak{g}^\perp = \{x \in \mathfrak{g} : K(x,y) = 0,\ \forall y \in \mathfrak{g}\}.$$

This is an ideal, to show why: take $x \in \mathfrak{g}^\perp$ and $y, z \in \mathfrak{g}$. Then we need to show that

$$K([x,y],z]) = 0.$$

But

$$K([x,y],z) = K(x,[y,z]) = 0$$

since $x \in \mathfrak{g}^\perp$.

Lemma 7.15. *Let I be an ideal of $\mathfrak{g}$ and let K_I be the killing form of I. Then*

$$K_I(x,y) = K(x,y),\ \forall x, y \in I.$$

Proof. Choose a basis of I and extend to a basis for $\mathfrak{g}$. Let $x, y \in I$, then with respect to this basis:

$$\operatorname{ad}(x) = \begin{pmatrix} A & * \\ 0 & 0 \end{pmatrix}$$

where $A = \operatorname{ad}(x)|_I$. And similarly for $\operatorname{ad}(y)$. Then we have

$$K_I(x,y) = \operatorname{tr}(AB) = \operatorname{tr}(\operatorname{ad}(x), \operatorname{ad}(y)) = K(x,y).$$

Theorem 7.16 (Cartan's criterion (for solvability))**.** *Suppose $\mathfrak{g}$ is a subalgebra of $\mathfrak{gl}(V)$, for V a finite dimensional $\mathbb{C}$-space. If $K(x,y)$ is identically zero on $\mathfrak{g} \times [\mathfrak{g},\mathfrak{g}]$ then $\mathfrak{g}$ is solvable. (This is saying that $\mathfrak{g}^{(1)} \leqslant \mathfrak{g}^\perp$).*

Proof. A big linear algebra hack, in Humphreys 4.3, uses Lie's theorem and Jordan decomposition (which we will cover next chapter). Not a very hard proof[5], just very long. This fails in $\operatorname{char} k = p$ as well. □

Corollary 7.17.

1. *If $\mathfrak{g} = \mathfrak{g}^\perp$ then $\mathfrak{g}$ is solvable.*
2. *If $\mathfrak{g}$ is simple, then $\mathfrak{g}^\perp = 0$*
3. *$\mathfrak{g}^\perp$ is solvable for any finite dimensional $\mathfrak{g}$.*

Proof.

[5] apparently

1. Consider the adjoint
$$\mathrm{ad} : \mathfrak{g} \to \mathfrak{gl}(V)$$
The image $\mathrm{ad}(\mathfrak{g})$ is $\mathfrak{g}$ modulo the kernel, so
$$\mathrm{ad}(\mathfrak{g}) = \frac{\mathfrak{g}}{Z(\mathfrak{g})}.$$
$Z(\mathfrak{g})$ is solvable since it's abelian. Since $\mathfrak{g} = \mathfrak{g}^{\perp}$, Cartan's criterion (7.16) implies $\mathrm{ad}(\mathfrak{g})$ is solvable. So by (7.9), $\mathfrak{g}$ must be solvable.
2. $\mathfrak{g}^{\perp}$ is an ideal, so either $\mathfrak{g}^{\perp} = 0$ (in which case we're done) or $\mathfrak{g}^{\perp} = \mathfrak{g}$, but then $\mathfrak{g}$ is solvable by what we've just proved, which contradicts the fact that $\mathfrak{g}$ is simple.
3. $(\mathfrak{g}^{\perp})^{\perp} = \mathfrak{g}^{\perp}$ by (7.15) and basic properties of this ideal, so by 1. which we just proved, $\mathfrak{g}^{\perp}$ is solvable.

□

Proof of (7.13).

(2) ⇒ (3): $\mathfrak{g}$ is a solvable ideal in its own right, so $\mathfrak{g}^{\perp} \subseteq \mathrm{Rad}(\mathfrak{g})$ which is assumed to be 0, so $\mathfrak{g}^{\perp} = 0$

(3) ⇒ (2): Let A be an abelian ideal of $\mathfrak{g}$. We claim $A \subseteq \mathfrak{g}^{\perp}$. Let $x \in A, y \in \mathfrak{g}$. Choose a basis for A and extend to $\mathfrak{g}$. Then
$$\mathrm{ad}(x) = \begin{pmatrix} 0 & * \\ 0 & 0 \end{pmatrix}$$
$$\mathrm{ad}(y) = \begin{pmatrix} * & * \\ 0 & * \end{pmatrix}$$
So $\mathrm{tr}(\mathrm{ad}(x)\,\mathrm{ad}(y)) = 0$, therefore $A = 0$. And if $\mathrm{Rad}(\mathfrak{g}) \neq 0$ then $\mathfrak{g}^{\perp} \neq 0$.

(2) & (3) ⇒(1): If $\mathfrak{g}$ is simple, then we're done. If not, then choose some minimal non-trivial ideal I. Let
$$\mathfrak{g}_I = \{x \in \mathfrak{g} \,:\, K(x, y) = 0 \,\forall y \in I\}$$
This is an ideal of $\mathfrak{g}$ (prove this).

Claim 1: $\mathfrak{g} = I \oplus \mathfrak{g}_I$.

Since I is simple (by minimality) and being non-abelian by (2), we know that
$$I \cap \mathfrak{g}_I \subseteq I^{\perp} = 0.$$
Consider the map
$$\mathfrak{g} \xrightarrow{\simeq} \mathfrak{g}^* \xrightarrow{\text{restrict}} I^*$$

$$x \to K(x, \cdot) \to K|_{I^*}(x, \cdot).$$

The kernel of this map is $\mathfrak{g}_I$, which prove the claim. We then repeat this argument with g_I (choosing some minimal ideal of g_I). We can do this because (Exercise:) any ideal of $\mathfrak{g}_I$ must be an ideal of $\mathfrak{g}$ and so $\mathrm{Rad}(\mathfrak{g}_I) = 0$.

Claim 2: $(g_I)^\perp = 0$. Since if $\alpha \in (g_I)^\perp$ then $x \in \mathfrak{g}^\perp$

(1) $\Rightarrow$ (2) Write $\mathfrak{g} = \oplus I_j$, I_j simple ideals. Let π_j be projection onto I_j.

□

Lecture 7

Exercise: If J is an ideal of $\mathfrak{g}$, then $\pi_j(J)$ is an ideal of I_j

Now if $A \subseteq \mathfrak{g}$ is an abelian ideal of $\mathfrak{g}$, then $\pi_j(A)$ is an abelian ideal of I_j, so $\pi_j(A) = 0 \, \forall j$ so $A = 0$.

Theorem 7.18 (Weyl)**.** *Any finite dimensional representation of a semisimple Lie algebra is completely reducible.*

Idea. Remember in the $\mathfrak{sl}_2$ case, where we used the Casimir element, there is a general idea of a Casimir element for a general semisimple Lie algebra. This is how we prove this claim.

Proof. The main ingredient is this new generalised Casimir element.

Exercise: Any ideal or quotient of a semisimple Lie algebra is also semisimple.

Definition 7.19 (General Casimir element)**.** Take any representation

$$\phi : \mathfrak{g} \to \mathfrak{gl}(V)$$

a finite dimensional irreducible representation of a semisimple Lie algebra $\mathfrak{g}$. Then without loss of generality, assume that ϕ is faithful, otherwise just work with $\frac{\mathfrak{g}}{\ker \phi}$. By Cartan-Killing criterion, $(-,-)$ the trace form, is non degenerate. Pick basis $x_1, \ldots, x_n$ of $\mathfrak{g}$. Since the trace form is non-degenerate, we can also pick a dual basis $y_1, \ldots, y_n$. Then let

$$\Omega_\phi = \sum \phi(x_i)\phi(y_i).$$

In Humphreys' book, this is called *the* **Casimir element** of ϕ. Note

- Ω_ϕ is an endomorphism of V
- Ω_ϕ commutes with $\phi(x)$, $\forall x \in \mathfrak{g}$.

By Schur's Lemma, Ω_ϕ is a scalar multiple of the identity on V, and $\operatorname{tr}\big(\Omega_{\phi)}\big) = \sum_i \operatorname{tr}(\phi(x_i)\phi(y_i)) = \dim \mathfrak{g}$ and it's also clear that Ω_ϕ is independent of the basis of $\mathfrak{g}$ which we chose.

Example. Let $\mathfrak{g} = \mathfrak{sl}_2 \leqslant \mathfrak{gl}_2$, with $V = \mathbb{C}^2$, and $\phi : \mathfrak{g} \to \mathfrak{gl}(V)$ is the defining representation of $\mathfrak{sl}_2$, then take $\{e, f, h\}$, the standard basis of $\mathfrak{sl}_2$. The dual basis relative to the trace form is $\{f, \frac{1}{2}h, e\}$. Then

$$\Omega_\phi = ef + \frac{1}{2}h^2 + fe = \begin{pmatrix} \frac{3}{2} & 0 \\ 0 & \frac{3}{2} \end{pmatrix}.$$

The $\frac{3}{2}$ comes from 3 being the dimension of the algebra, and 2 being the dimension of the representation. The proof of Weyl's theorem follows the same as for $\mathfrak{sl}_2$, but with this general casimir element.

8 Jordan decomposition

First, two observations:

- If $\mathfrak{g}$ is a simple Lie algebra, and $\phi : \mathfrak{g} \to \mathfrak{gl}(V)$ is a finite dimensional representation, then $\phi(\mathfrak{g}) \subseteq \mathfrak{sl}(V)$. This is because $[\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}$, so

$$\phi(g) = \phi([\mathfrak{g}, \mathfrak{g}]) = [\phi(\mathfrak{g}), \phi(\mathfrak{g})] \subseteq [\mathfrak{gl}(V), \mathfrak{gl}(V)] = \mathfrak{sl}(V).$$

- Recall that if $x \in \mathfrak{gl}(V)$, then there is a basis of V such that x is a block diagonal, with Jordan blocks of form:

$$\begin{pmatrix} \lambda & 1 & 0 & \dots & 0 \\ 0 & \lambda & 1 & \dots & 0 \\ 0 & 0 & \ddots & \ddots & 0 \\ 0 & 0 & 0 & \ddots & 1 \\ 0 & 0 & 0 & 0 & \lambda \end{pmatrix}$$

Definition 8.1 (Nilpotent/Semisimple elements)**.** We say a element x is **nilpotent** if $x^n = 0$ for some n and we say an element is **semisimple** if the roots of its minimal polynomial are distinct (i.e. it is diagonalisable).

Proposition 8.2 (Jordan Decomposition)**.** *If $x \in \mathfrak{gl}(V)$, and V finite dimensional, then*

1. *There are unique $x_s, x_n \in \mathfrak{gl}(V)$ with x_s semisimple, x_n nilpotent, and $x = x_n + x_s$ and $[x_n, x_s] = 0$.*

2. *There exist polynomials $p_s, p_n \in \mathbb{C}[T]$ without constant terms such that $x_s = p_s(x)$, $x_n = p_n(x)$ and x_s, x_n commute with anything which commutes with x.*

3. *If $A \leqslant B \leqslant V$ are subspaces, and $x : B \to A$, then x_s and x_n also map $B \to A$.*

The decomposition $x = x_n + x_s$ is called the (additive) Jordan(-Chevalley) decomposition of x. x_n and x_s are called the nilpotent part/semisimple part of x respectively.

Proof. Routine Linear Algebra (Done in Humphreys). □

Example. If x is represented by a single Jordan block, then

$$x_s = \begin{pmatrix} \lambda & 0 & 0 & 0 \\ 0 & \lambda & 0 & 0 \\ 0 & 0 & \ddots & 0 \\ 0 & 0 & 0 & \lambda \end{pmatrix}$$

and

$$x_n = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & \ddots & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Why is this a valuable thing to know?

Suppose $\mathfrak{g}$ is a finite dimensional Lie algebra. Then we have an adjoint representation

$$\mathrm{ad} : \mathfrak{g} \to \mathfrak{gl}(V)$$

for V finite dimensional. Then if $x \in \mathfrak{gl}(V)$ is nilpotent, so is $\mathrm{ad}(x)$ (prove this). Similarly, if x is semisimple, then so is $\mathrm{ad}(x)$ (prove this too).

Lemma 8.3. *Let $x \in \mathfrak{g} \leqslant \mathfrak{gl}(V)$ with $\dim V < \infty$. Then $x = x_s + x_n$ is its Jordan decomposition. Then*

$$\mathrm{ad}(x) = \mathrm{ad}(x_s) + \mathrm{ad}(x_n)$$

is the Jordan decomposition of $\mathrm{ad}(x)$ in $\mathfrak{gl}(V)$.

Proof. $\mathrm{ad}(x_s)$ and $\mathrm{ad}(x_n)$ are respectively semisimple and nilpotent. They commute since

$$[\mathrm{ad}\, x_s, \mathrm{ad}\, x_n] = \mathrm{ad}([x_s, x_n]) = 0$$

By uniqueness in (8.2) the Jordan decomposition must be $\operatorname{ad} x = \operatorname{ad} x_s + \operatorname{ad} x_n$. □

Theorem 8.4. *Suppose $\mathfrak{g}$ is semisimple and a subalgebra of $\mathfrak{gl}(V)$ and $x \in \mathfrak{g}$. Then $x_s, x_n \in \mathfrak{g}$.*

Proof. Let

$$N(\mathfrak{g}) = \{y \in \mathfrak{gl}(V) : [y, z] \in \mathfrak{g} \ \forall z \in \mathfrak{g}\},$$

be called the **normalizer** of $\mathfrak{g}$ in $\mathfrak{gl}(V)$. We claim that

1. $N(\mathfrak{g})$ is a subalgebra of $\mathfrak{gl}(V)$ which contains $\mathfrak{g}$ as an ideal.
2. x_s and x_n are in $N(\mathfrak{g})$

Claim 1 is clear by the definition of $N(\mathfrak{g})$.

For 2, we have that for $z \in \mathfrak{g}$ we have $[x_s, z] = \operatorname{ad}(x_s)(z) = \operatorname{ad}(x)_s(z)$ and (by 8.2 part 2), this is in $\mathfrak{g}$, since it is a polynomial in $\operatorname{ad}(x)$ with no constant term.

Let W be an irreducible subrepresentation of V, and define

$$\mathfrak{g}_W = \{y \in \mathfrak{gl}(V) : yw \in W \ \forall w \in W, \text{ and } \operatorname{tr}(y)_W = 0\}.$$

Where $\operatorname{tr}(y)_W$ is the trace of the restriction of y to W. Then we claim that

$$\mathfrak{g} \leqslant \mathfrak{g}_W.$$

W is a subrepresentation, therefore W must be stabilised by $\mathfrak{g}$. The image of $\mathfrak{g}$ in $\mathfrak{gl}(W)$, call this $\bar{\mathfrak{g}}$, must also be stabilised, then this is also semisimple.

Therefore

$$[\bar{\mathfrak{g}}, \bar{\mathfrak{g}}] = \bar{\mathfrak{g}},$$

therefore every element of $\bar{\mathfrak{g}}$ is a sum of commutators (where the trace of every commutator must be 0 by the definition of $\mathfrak{g}_W$). Therefore $\mathfrak{g} \leqslant \mathfrak{g}_W$. Thus $\operatorname{tr}(y|_W) = 0$. Note x_s, x_n are polynomials in x, therefore, if x stabilises something then x_s and x_n must stabilise it as well. Also $\operatorname{tr}(x_n|_W) = 0$, since $x_n|_W$ is nilpotent. Then

$$\operatorname{tr}(x_s|_W) = \operatorname{tr}(x|_W - x_n|_W) = \operatorname{tr}(x|_W) - \operatorname{tr}(x_n|_W) = 0.$$

Therefore $x_s, x_n \in \mathfrak{g}_W$.

Then to finish, we need to show that x_s and x_n are in $\mathfrak{g}$. Define

$$\mathfrak{g}' = N(\mathfrak{g}) \cap \left(\bigcap_{W \leqslant V,\ W \text{ irred}} \mathfrak{g}_W \right).$$

And we claim $\mathfrak{g} = \mathfrak{g}'$, so that since x_s and x_n are in all of the $\mathfrak{g}_W$ in the intersection, they must also lie in $\mathfrak{g}'$ and thus in $\mathfrak{g}$.

Since $\mathfrak{g}' \leqslant N(\mathfrak{g})$, $\mathfrak{g}$ is an ideal of $\mathfrak{g}'$. Then $\mathfrak{g}$ is a subrepresentation of $\mathfrak{g}'$ under the adjoint representation of $\mathfrak{g}$. By Weyl's theorem (7.18) applied to $\mathfrak{g}'$ [6], $\mathfrak{g}' = \mathfrak{g} \oplus U$ as representations of $\mathfrak{g}$ for some subrepresentation U.

Claim: $U = 0$. Take $u \in U$. We have $[u, \mathfrak{g}] \subseteq \mathfrak{g}$ as $\mathfrak{g}$ is an ideal of $\mathfrak{g}'$. But $\operatorname{ad} \mathfrak{g} U \leqslant U$[7], so $[u, \mathfrak{g}] \subseteq U \cap \mathfrak{g} = 0$.

Therefore u commutes with every element of $\mathfrak{g}$. Hence u is a $\mathfrak{g}$-homomorphism $V \to V$. So u stabilises every irreducible subrepresentation W, so by Schur's lemma, we have

$$u|_W = \lambda \operatorname{Id}_W$$

for some scalar λ. Now $\operatorname{tr}(u|_W) = 0$ for all irreducible $W \leqslant V$ because $u \in \mathfrak{g}_W$ for each W. Therefore

$$\operatorname{tr}(\lambda \operatorname{Id}_W) = 0 \Rightarrow \lambda = 0 \Rightarrow u|_W = 0.$$

Finally,

$$V \cong \bigoplus_i W_i \quad \text{some irreducibles } W_i.$$

Therefore $u = 0$. □

For $\mathfrak{g}$ as in statement, we can define an "abstract" Jordan decomposition by

$$\operatorname{ad}(x) = \operatorname{ad}(x)_s + \operatorname{ad}(x)_n.$$

As ad is faithul, for $\mathfrak{g}$ semisimple, we have

$$\mathfrak{g} \cong \operatorname{ad} \mathfrak{g} \leqslant \mathfrak{gl}(\mathfrak{g}).$$

By (8.4) $\operatorname{ad}(x)_s, \operatorname{ad}(x)_n \in \operatorname{ad}(\mathfrak{g})$, hence are of the form $\operatorname{ad}(x_s), \operatorname{ad}(x_n)$ for some elements $x_s, x_n \in \mathfrak{g}$.

$$\operatorname{ad} x = \operatorname{ad}(x_s + x_n) \Rightarrow x = x_s + x_n$$

by faithfulness.

Suppose $\mathfrak{g} \leqslant \mathfrak{gl}(V)$, for some V. Write $x = x_s + x_n$. By (8.3), $\operatorname{ad}(x_s) = \operatorname{ad}(x)_s$, $\operatorname{ad}(x_n) = \operatorname{ad}(x)_n$. So the "abstract" decomposition we just gave agrees with the original definition.

[6] We know it is semisimple since we have a homomorphism to it from a semisimple Lie algebra, thus we can apply Weyl's theorem.

[7] since the adjoint acts faithfully.

Corollary 8.5. *Let a representation*

$$\varphi : \mathfrak{g} \to \mathfrak{gl}(V)$$

be any representation of a semisimple Lie algebra $\mathfrak{g}$*. Let* $x \in \mathfrak{g}$ *have Jordan decomposition* $x = x_s + x_n$*. Then*

$$\varphi(x)_s = \varphi(x_s)$$

and

$$\varphi(x)_n = \varphi(x_n)$$

defines a Jordan decomposition of $\varphi(x)$*.*

Proof. Omitted. [8]

End of Introduction

There is a section 9, but it rears its head later. On to section 10.

10 Cartan subalgebras and root space decompositions

Here, as usual, $\mathfrak{g}$ is some finite dimensional, semisimple Lie algebra over $\mathbb{C}$.

Definition 10.1 (Toral/Cartan subalgebra)**.** A subalgebra $\mathfrak{t} \subseteq \mathfrak{g}$ is **toral** if

1. $\mathfrak{t}$ is abelian.
2. $\operatorname{ad} x$ is semisimple for all $x \in \mathfrak{t}$.

A maximal toral subalgebra is called a maximal torus or a **Cartan subalgebra** (CSA).

Remark. Many authors, including Humphreys, define Cartan subalgebras as nilpotent subalgebras which are equal to their normalizers in $\mathfrak{g}$. This is equivalent to the definition above.

Examples.

1. $\mathfrak{g} \leqslant \mathfrak{sl}_n, \mathfrak{gl}_n$, then $\mathfrak{t} = \{\text{diagonal matrices in } \mathfrak{g}\}$ is a maximal torus.
2. Similarly for $\mathfrak{so}_n, \mathfrak{sp}_{2n}$.

[8] In the 2015 notes by Stewart. It is his "Corollary 5.11". *Needs* semisimplicity, *definitely needs* to be over $\mathbb{C}$.

Lemma 10.2. *Let $t_1, \ldots, t_n : V \to V$ be pairwise commuting ($t_i t_j = t_j t_i$) semisimple endomorphisms of V (f.d. over $\mathbb{C}$). Let*

$$\lambda = (\lambda_1, \ldots, \lambda_n) \in \mathbb{C}^n.$$

Then set

$$V_\lambda = \{v \in V : t_i(v) = \lambda_i v, \ \forall 1 \leqslant i \leqslant n\}$$

(the simultaneous eigenspaces for all t_i). Then

$$V = \bigoplus_{\lambda \in \mathbb{C}^n} V_\lambda.$$

Proof. Induction on n.

$\underline{n = 1}$:

This follows since t_1 is semisimple, so the diagonalisability means that V has a basis of t_1-eigenvectors, so decomposes as t_1-eigenspaces.

$\underline{n > 1}$:

We know by induction

$$V = \bigoplus_{\lambda' \in \mathbb{C}^{n-1}} V_{\lambda'}$$

for the action of $t_1, \ldots, t_{n-1}$. Then since t_i commute,

$$t_n(V_{\lambda'}) = V_{\lambda'} \ \forall \lambda'.$$

So decomposing each $V_{\lambda'}$ for t_n (exactly as in the $n = 1$ case), we're done. □

Lemma 10.3. *Any $\mathfrak{g}$ contains a Cartan subalgebra.*

Proof. We need something called Engel's[9] theorem, which is on sheet 2 as a further problem. We also need Zorn's lemma.

Recasting (10.2), suppose $\mathfrak{h} \leqslant \mathfrak{gl}(V)$ with a basis of commuting semisimple elements $t_1, \ldots, t_n$. Then when we write $\lambda \in \mathbb{C}^n$, we mean it to correspond to the element of $\mathfrak{h}^*$ (the dual) given by

$$t_i \mapsto \lambda_i.$$

[9] Not that one, also he's called Engel, not Engels. However, they do both have first name Friedrich.

Then

$$V_\lambda = \{v \in V \,:\, hv = \lambda(h)v \ \forall h \in \mathfrak{h}\}.$$

In our situation, fix $\mathfrak{t} \leqslant \mathfrak{g}$ a Cartan subalgebra, then

$$\mathfrak{g} = \bigoplus_{\lambda \in \mathfrak{t}^*} \mathfrak{g}_\lambda,$$

where

$$\mathfrak{g}_\lambda = \{x \in \mathfrak{g} \,:\, [t, x] = \lambda(t)x, \ \forall t \in \mathfrak{t}\}.$$

□

Definition 10.4 (Roots/Root Space)**.** Let

$$\Phi = \{\alpha \in \mathfrak{t}^* \setminus \{0\} \,:\, \mathfrak{g}_\alpha \neq 0\}.$$

The elements of Φ are the **roots** of $\mathfrak{g}$ with respect to $\mathfrak{t}$. If $\alpha \in \Phi$, $\mathfrak{g}_\alpha$ is called a **root space** and

$$\mathfrak{g} = \mathfrak{g}_0 \oplus \sum_{\alpha \in \Phi} \mathfrak{g}_\alpha.$$

is the **root space decomposition** or **Cartan decomposition** of $\mathfrak{g}$.

Proposition 10.5.

1. *For all* $\alpha, \beta \in \ \mathfrak{t}^*$, $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] \subseteq \mathfrak{g}_{\alpha+\beta}$.
2. *If* $\alpha \in \Phi$, *then if* $x \in \mathfrak{g}_\alpha$, $\mathrm{ad}(x)$ *is nilpotent.*
3. *If* $\alpha + \beta \neq 0$, $K(\mathfrak{g}_\alpha, \mathfrak{g}_\beta) = 0$ *for all* $\alpha, \beta \in \mathfrak{t}^*$

Proof.

1. Take $x \in \mathfrak{g}_\alpha$, $y \in \mathfrak{g}_\beta$, $t \in \mathfrak{t}$. We have from the Jacobi identity that

$$\begin{aligned}
[t, [x, y]] &= -[x, [y, t]] - [y, [t, x]] \\
&= [x, [t, y]] - \alpha(t)[y, x] \\
&= \beta(t)[x, y] + \alpha(t)[x, y] \\
&= (\alpha + \beta))(t)[x, y]
\end{aligned}$$

2. Use 1, and the finite dimensionality of $\mathfrak{g}$.
3. If $\alpha + \beta \neq 0$, then have that

$$\exists t \in \mathfrak{t} \text{ such that } (\alpha + \beta)(t) \neq 0.$$

So then fix such a t and fix $x \in \mathfrak{g}_\alpha$, $y \in \mathfrak{g}_\beta$. Then

$$\begin{aligned}\alpha(t)K(x,y) &= K([t,x],y)\\ &= -K([x,t],y)\\ &= -K(x,[t,y])\\ &= -\beta(t)K(x,y)\end{aligned}$$

Which implies $(\alpha+\beta)(t)K(x,y) = 0$. Therefore $K(x,y) = 0$.

□

Corollary 10.6.

1. *$K|_{\mathfrak{g}_0 \times \mathfrak{g}_0}$ is non-degenerate.*
2. *If $\alpha \in \Phi$, then $-\alpha \in \Phi$.*

Proof.

1. If $z \in \mathfrak{g}_0$ and $K(z,y) = 0$ for all $y \in \mathfrak{g}_0$, then by (10.5(3)) $\mathfrak{g}_0$ is orthogonal to $\mathfrak{g}_\alpha$ for $\alpha \neq 0$. If $x \in \mathfrak{g}$ then we can write $x = x_0 + \sum_{\alpha \in \Phi} x_\alpha$ with $x_\alpha \in \mathfrak{g}_\alpha$. By linearity $K(z,x) = 0$ for all $x \in \mathfrak{g}$. Since K is non-degenerate on $\mathfrak{g}$, we must have $z = 0$.
2. Similar to above.

□

Proposition 10.7.

$$\mathfrak{g}_0 = \mathfrak{t}$$

Proof. Humphreys (8.2) □

Corollary 10.8. *$K|_{\mathfrak{t}\times\mathfrak{t}}$ is non-degenerate. In particular*

$$\begin{aligned}\mathfrak{t} &\to \mathfrak{t}^*\\ x &\mapsto K(x,\,\cdot\,)\end{aligned}$$

is an isomorphism.

Proof. Let inverse be $\lambda \mapsto t_\lambda$, then $t_\lambda \in \mathfrak{t}$ (called the coroot associated to λ is defined by

$$K(t_\lambda, x) = \lambda(x)$$

for all $x \in \mathfrak{t}$. □

Examples.

1. $\mathfrak{g} = \mathfrak{sl}_2$,
$$\mathfrak{t} = \left\langle \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = h \right\rangle$$
Define $\alpha \in \mathfrak{t}^*$ by $\alpha(h) = 2$. Then $\mathfrak{g}_\alpha = \langle e \rangle$, $\mathfrak{g}_{-\alpha} = \langle f \rangle$ and $\mathfrak{sl}_2 = \mathfrak{t} \oplus \mathfrak{g}_\alpha \oplus \mathfrak{g}_{-\alpha}$.

2. Let $\mathfrak{g} = \mathfrak{sl}_3$. Take
$$h_1 = \begin{pmatrix} 1 & & \\ & -1 & \\ & & 0 \end{pmatrix}$$
and
$$h_2 = \begin{pmatrix} 0 & & \\ & 1 & \\ & & -1 \end{pmatrix}$$
and let $\mathfrak{t} = \langle h_1, h_2 \rangle$. Let $\alpha_i \in \mathfrak{t}^*$ be such that $\alpha_i(\mathrm{Diag}(a_1, a_2, a_3)) = a_i$ Then
$$\mathfrak{sl}_3 = \mathfrak{t} \oplus \mathfrak{g}_{\alpha_1 - \alpha_2} \oplus \mathfrak{g}_{\alpha_1 - \alpha_3} \oplus \mathfrak{g}_{\alpha_2 - \alpha_1} \oplus \mathfrak{g}_{\alpha_2 - \alpha_3} \oplus \mathfrak{g}_{\alpha_3 - \alpha_1} \oplus \mathfrak{g}_{\alpha_3 - \alpha_2}$$
where $\mathfrak{g}_{\alpha_i - \alpha_j} = \langle e_{ij} \rangle$. Try decomposing the adjoint this way.

3. Try $\mathfrak{g} = \mathfrak{sl}_n$, $\mathfrak{t} =$ diagonals in $\mathfrak{g}$, then try $\mathfrak{g} = \mathfrak{so}_{2n}, \mathfrak{so}_{2n+1}, \mathfrak{sp}_{2n}$.

Proposition 10.9. *If $\alpha \in \Phi$ is a root, $e_\alpha \in \mathfrak{g}_\alpha$, then $\exists f_\alpha \in \mathfrak{g}_{-\alpha}$ such that*
$$\mathfrak{m}_\alpha = \langle e_\alpha, f_\alpha, h_\alpha \rangle = [e_\alpha, f_\alpha] \cong \mathfrak{sl}_2.$$
We're saying that every semisimple Lie algebra is "made out of" $\mathfrak{sl}_2$'s.

First, note that if $t \in \mathfrak{t}$ satsifies $\alpha(t) = 0$ for all $\alpha \in \Phi$, then $t = 0$. Since, if $\alpha \in \Phi$ and $x \in \mathfrak{g}_\alpha$ then
$$0 = \alpha(t)(x) = [t, x].$$
Since a toral subalgebra is abelian, this holds in all of $\mathfrak{g}$. $t \in Z(\mathfrak{g}) = \{0\}$. This gives us the following lemma.

Lemma 10.10. *Φ spans $\mathfrak{t}^*$*

Proof. If not $\exists t \neq 0$ such that $\alpha(t) = 0$ for all $\alpha \in \Phi$ which is false. □

Proof of 10.9.

Claim 1: $[\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}]$ is 1-dimensional.

Take $x \in \mathfrak{g}_\alpha$, $y \in \mathfrak{g}_{-\alpha}$, so $[x, y] \in \mathfrak{t}$. Let $t \in \mathfrak{t}$. So

$$K([x, y], t) = K(x, [y, t]) = -K(x, [t, y]) = \alpha(t)K(x, y)$$

Therefore $[x, y] = K(x, y)t_\alpha \in \langle t_\alpha \rangle$. Therefore we know that $\dim[\mathfrak{g}_{\alpha,\mathfrak{g}_{-\alpha}}] \leqslant 1$, but then

$$\exists x, y \text{ such that } K(x, y) \neq 0$$

since K is non-degenerate. Therefore the dimension must be 1.

Claim 2: $\alpha(t_\alpha) \neq 0$.

Since the Killing form is non-degenerate, and rescaling is possible, we may choose $x \in \mathfrak{g}_\alpha$ and $y \in \mathfrak{g}_{-\alpha}$ such that $K(x, y) = 1$. Then

$$[x, y] = K(x, y)t_\alpha = t_\alpha$$

and we have

$$[t_\alpha, x] = \alpha(t_\alpha)x, \ [t_\alpha, y] = -\alpha(t_\alpha)y.$$

Therefore $\langle \{x, y, t_\alpha\} \rangle$ is a subalgebra, say $\mathfrak{h}$, of $\mathfrak{g}$. Assume that $\alpha(t_\alpha) = 0$. Since

$$[\mathfrak{h}, \mathfrak{h}] = \langle t_\alpha \rangle,$$

this means that $\mathfrak{h}$ is a solvable algebra. Now

$$\mathrm{ad} : \mathfrak{g} \to \mathfrak{gl}_\mathfrak{g},$$

and we know $\mathfrak{h}$ embeds into a solvable subalgebra of $\mathfrak{gl}(\mathfrak{g})$, with respect to which

$$\mathrm{ad}(\mathfrak{h}) \subseteq \{\text{Upper triangular matrices}\},$$

and $\mathrm{ad}(t_\alpha) = [\mathrm{ad}\, x, \mathrm{ad}\, y]$ is a stricly upper triangular matrix. Consider $\mathrm{ad}(t_\alpha)$ is nilpotent, but then by semisimplicity (of the algebra), $\mathrm{ad}(t_\alpha)$ is semisimple. Therefore

$$\mathrm{ad}(t_\alpha = 0.$$

Therefore $t_\alpha \in Z(\mathfrak{g}) = \{0\}$.

Claim 3: $[[\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}], \mathfrak{g}_\alpha] \neq 0$.

If $x \in \mathfrak{g}_\alpha, y \in \mathfrak{g}_{-\alpha}$ with $K(x, y) \neq 0$, then for all $z \in \mathfrak{g}_\alpha$:

$$[[x, y], z] = [K(x, y)t_\alpha, z] = K(x, y)\alpha_{t_\alpha} z,$$

which is non-zero if $z \neq 0$.

Lecture 8

Claim 4: If $\alpha \in \Phi$, $e_\alpha \in \mathfrak{g}_\alpha$, then $\exists f_\alpha \in \mathfrak{g}_{-\alpha}$ such that

$$\mathfrak{m}_\alpha = \langle e_\alpha, f_\alpha, h_\alpha = [e_\alpha, f_\alpha] \rangle \cong \mathfrak{sl}_2.$$

Take an element $e_\alpha \in \mathfrak{g}_\alpha$ in the root space, and find $f_\alpha \in \mathfrak{g}_{-\alpha}$ such that

$$K(e_\alpha, f_\alpha) = \frac{2}{\alpha(t_\alpha)}.$$

Then define

$$h_\alpha = \frac{2}{K(t_\alpha, t_\alpha)}.$$

Check that the $\mathfrak{sl}_2$ relations hold:

$$[e_\alpha, f_\alpha] = K(e_\alpha, f_\alpha) t_\alpha = h_\alpha$$
$$[h_\alpha, e_\alpha] = \frac{2}{\alpha(t_\alpha)} [t_\alpha, e_\alpha] = 2e_\alpha.$$

and similarly

$$[h_\alpha, f_\alpha] = -2f_\alpha.$$

So this is indeed isomorphic to $\mathfrak{sl}_2$. The set $\{e_\alpha, f_\alpha, h_\alpha\}$ is called a $\mathfrak{sl}_2$-triple.

Exercise: Show that "weights add": if $\mathfrak{g}$ is semisimple with root space decomposition

$$\mathfrak{g} = \mathfrak{t} \oplus \bigoplus_\alpha \mathfrak{g}_\alpha,$$

and V, W are representations of $\mathfrak{g}$, which therefore have corresponding weight spaces V_α and W_α. Then

1. $\mathfrak{g}_\alpha V_\beta \subseteq V_{\alpha+\beta}$.
2. $V_\alpha \otimes W_\beta \subseteq (V \otimes W)_{\alpha+\beta}$.

Lemma 10.11.

1. *If V is a finite dimensional representation for $\mathfrak{g}$, then $V|_{\mathfrak{m}_\alpha}$ is a finite dimensional representation.*
2. *If $\beta \in \Phi$ or $\beta = 0$, then let*

$$V = \oplus \mathfrak{g}_{\beta + c\alpha},$$

where the direct sum is taken over all $c \in \mathbb{C}$ such that $\beta + c\alpha \in \Phi$. This is a representation of $\mathfrak{m}_\alpha$ via restriction of the adjoint.

Proof.

1. Clear by generic facts about representations and restrictions.
2. Clear from (10.5)

□

Definition (Root string)**.** We call V as obtained in Lemma 10.11 the **α-root string through** β.

Proposition 10.12. *Let $\alpha \in \Phi$. The root spaces $\mathfrak{g}_{\pm\alpha}$ are 1-dimensional. Moreover the multiples of α which lie in Φ are $\pm\alpha$.*

Proof. If $c\alpha \in \Phi$, then h_α takes $c\alpha(h_\alpha) = 2c$ as an eigenvalue. The eigenvalues of h_α are integral. Either $c \in \mathbb{Z}$ or $c \in \mathbb{Z} + \frac{1}{2}$.

Let

$$V = \mathfrak{t} \oplus \bigoplus_{c\alpha \in \Phi} \mathfrak{g}_{c\alpha}.$$

Let $K = \ker \alpha \leqslant \mathfrak{t}^*$. It is easy to check that $K \oplus \mathfrak{m}_\alpha$ is a $\mathfrak{m}_\alpha$ subrepresentation of V[10]. By Weyl's theorem, as a representation of $\mathfrak{m}_\alpha$, V must split as

$$V = K \oplus \mathfrak{m}_\alpha \oplus W,$$

where W is a complementary subrepresentation of V. Assume either of the conclusions in the statement of the proposition are false, then $W \neq 0$. Let $W_0 (\cong V(s))$ be an irreducible representation of W. Then W_0 has a highest weight vector $w_0 \in \mathfrak{g}_{c\alpha}$ for some $c \in \mathbb{C}$. Then

$$[h_\alpha, w_0] = sw_0,$$

for some s. We now proceed by cases:

Case 1: If s is even, then 0 must be an eigenvalue for the action of h_α (since eigenvalues go down in multiples of 2). Call e the eigenvector. The 0-eigenspace of h_α on V is $\mathfrak{t}$, which is contained in $K \oplus \mathfrak{m}_\alpha$. So $v \in (K \oplus \mathfrak{m}_\alpha) \cap W_0 = 0$. This is a contradiction, eigenvectors cannot be 0.

An aside: If $2\alpha \in \Phi$, then h_α has $2\alpha(h_\alpha) = 4$ as an eigenvalue. The eigenvalues of h_α on $K \oplus \mathfrak{m}_\alpha$ are 0 and ± 2. So the only way this could happen is if W contains an irreducible subrepresentation $V(s)$ where s is even (which we just proved can't happen). "Twice a root is never a root."

[10] Since $\mathfrak{m}_\alpha$ acts trivially on K, so when we take the direct sum we still get a representation.

Case 2: s odd. h_α will act with an eigenvalue 1, so has an eigenvector for this eigenvalue, say e.

As $\alpha(h_\alpha) = 2$, this means $\frac{1}{2}\alpha$ is a root of $\mathfrak{g}$. Now we have that $\frac{1}{2}\alpha$ and α are both roots of $\mathfrak{g}$. This is contrary to the aside. □

Exercise: Show $\mathfrak{m}_\alpha \cong \mathfrak{m}_{-\alpha}$ canonically[11], and $h_\alpha \cong -h_{-\alpha}$.

Proposition 10.13. *Take $\alpha, \beta \in \Phi$ such that $\beta \neq \pm\alpha$.*

1. *$\beta(h_\alpha) \in \mathbb{Z}$, the **Cartan Integers**.*
2. *There exist integers $p, q \geqslant 0$ such that if $r \in \mathbb{Z}m$ then $\beta + r\alpha \in \Phi \Leftrightarrow -p \leqslant r \leqslant q$. Moreover $p - q = \beta(h_\alpha)$.*
3. *$[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] = \mathfrak{g}_{\alpha+\beta}$.*

Proof.

1. Consider $V = \bigoplus_{r\in\mathbb{Z}} \mathfrak{g}_{\beta+r\alpha}$ and let $\mathfrak{m}_\alpha$ act on V by restriction (of the adjoint). Let $q = \max\{r : \beta + r\alpha \in \Phi\}$, and let $v \in \mathfrak{g}_{\beta+q\alpha}$ be non-zero. Then
$$[e_\alpha, v] \in \mathfrak{g}_{\beta+(q+1)\alpha} = 0,$$
and
$$[h_\alpha, v] = (\beta + q\alpha)(h\alpha)(v) = \langle v\rangle.$$
So v is a highest weight vector with weight $(\beta+q\alpha)(h_\alpha)$. Therefore, from what we know about the representation theory of $\mathfrak{sl}_2$ (i.e. h always acts by integers on eigenvectors), we know that
$$\beta(h_\alpha) + q\alpha(h_\alpha) \in \mathbb{Z}_{\geqslant 0}.$$
So, rewriting, we get
$$\beta(h_\alpha) + 2q \in \mathbb{Z}_{\geqslant 0}.$$
So we know that $\beta(h_\alpha) \in \mathbb{Z}$.
2. Recall (4.8), $W = \langle v, f_\alpha v, f_\alpha^2 v, \dots\rangle$ is an irreducible representation of V and h_α acts on it via the following matrix
$$\begin{pmatrix} (\beta+q\alpha)(h_\alpha) & & & \\ & (\beta+(q-1)\alpha)(h_\alpha) & & \\ & & \ddots & \\ & & & -(\beta_q\alpha)(h_\alpha) \end{pmatrix}$$

[11] They're both already isomorphic to $\mathfrak{sl}_2$

In particular, we have

$$W = \sum_{r=-p}^{q} \mathfrak{g}_{\beta+r\alpha}$$

for some p (namely, $p = \max\{r : \beta - r\alpha \in \Phi\}$). Suppose $W' \leqslant V$ is a subrepresentation not equal to W. Then W' contains a highest weight vector $w \in \mathfrak{g}_\gamma$, for some γ. Then $\gamma(h_\alpha) < -(\beta + q\alpha) \leqslant 0$, this is a contradiction since the highest weight vector must have eigenvalue greater than or equal to 0 when acted on by h_α. Finally

$$(\beta - p\alpha)(h_\alpha) = -(\beta + q\alpha)(h_\alpha),$$

i.e. $\beta(h_\alpha) = p - q$.

Lecture 9

3. $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] \subseteq \mathfrak{g}_{\alpha+\beta}$. So if $\mathfrak{g}_{\alpha+\beta} = 0$, then done. If not, then $\alpha + \beta \in \Phi$, so take $v \in \mathfrak{g}_\beta$ with $v \neq 0$. Then if $[e_\alpha, v] = 0$, then v is a HWV for V, contradiction since then it would generate a submodule of V, but V is irreducible. Therefore we must have that $[e_\alpha, v] \neq 0$ and $\mathfrak{g}_{\alpha+\beta}$ is then spanned by it.

□

Definition 10.14 (Reflection)**.** For $\alpha \in \Phi$, define the **reflection** w_α at α by:

$$\begin{aligned} w_\alpha : \mathfrak{t}^* &\to \mathfrak{t}^* \\ \beta &\mapsto \beta - \beta(h_\alpha)\alpha \end{aligned}$$

Corollary 10.15. $w_\alpha(\Phi) = \Phi$.

Proof. Let $\beta \in \Phi$, let p, q be as in 10.13. Then we need to show $\beta - \beta(h_\alpha)\alpha \in \Phi$. We have

$$\beta - \beta(h_\alpha)\alpha = \beta - (p - q)\alpha.$$

But, $-p \leqslant -(p - q) \leqslant q$, so $\beta - (p - q)\alpha$ must lie in the string of roots. □

w_α is a reflection over the **(root) hyperplane**, $H_\alpha = \{\lambda \in \mathfrak{t}^* : \lambda(h_\alpha) = 0\}$, and this reflection preserves Φ. We now define a **root system** as something having all the nice properties of Φ, and then show that there's a correspondence

$$\text{Root systems} \leftrightarrow \text{Semisimple Lie algebras.}$$

11 Root systems

There are two ways to think of root systems:

(a) Looking at roots in Euclidean space. Recall Φ spans $\mathfrak{t}^*$, from (10.10).

Proposition 11.1. *Define a bilinear form on* $\mathfrak{t}^*$ *by*

$$(\lambda, \mu) = K(t_\lambda, t_\mu)$$

for each $\lambda, \mu \in \mathfrak{t}^*$. *Then:*

1. *If* $\alpha, \beta \in \Phi$, *then* $(\alpha, \beta) \in \mathbb{Q}$.
2. *If we pick basis* $\alpha_1, \ldots, \alpha_l$ *of* $\mathfrak{t}^*$, *and* $\beta \in \Phi$, *then*

$$\beta = \sum c_i \alpha_i$$

   with $c_i \in \mathbb{Q}$. *That is to say,* $\dim_{\mathbb{Q}}(\Phi) = \dim_{\mathbb{C}}(\mathfrak{t})$.
3. $(\,\cdot\,,\,\cdot\,)$ *is positive definite on* $\mathbb{Q}\Phi = \operatorname{span}_{\mathbb{Q}}(\alpha : \alpha \in \Phi)$.

Proof. Omitted. [12]

(b) The other way of looking at root systems is **abstract root systems**.

Temporarily forget everything we just did, and approach from a different angle. We start with a Euclidean space:

$$(E, (\,\cdot\,,\,\cdot\,))$$

(A space with a positive definite, bilinear, symmetric form on it). If $0 \neq \alpha \in E$, define $\alpha^\vee : E \to \mathbb{R}$ by:

$$\alpha^\vee(\lambda) = \frac{2(\alpha, \lambda)}{(\alpha, \alpha)}.$$

Also define $w_\alpha : E \to E$ by

$$w_\alpha = \lambda - \alpha^\vee(\lambda)\alpha.$$

Definition 11.2 (Root system)**.** A finite subset $\Phi \subseteq E$ is a **root system** if

1. $0 \notin \Phi$ and Φ spans E.
2. If $\alpha, \beta \in \Phi$ then $\beta^\vee(\alpha) \in \mathbb{Z}$.
3. If $\alpha \in \Phi$, then $w_\alpha(\Phi) = \Phi$.
4. If $\alpha, c\alpha \in \Phi$ for c constant, then $c = \pm 1$.

[12] In Grojnowski's notes, proposition 4.7. For 1, observe that $\beta(h_\alpha) = \frac{2(\alpha,\beta)}{(\alpha,\alpha)}$.

Remark. Removing condition 4 gives a "non-reduced" root system (not of interest to us here).

Notation. If $\mu \in E$, and $\lambda \in E^*$, then we write

$$\langle \mu, \lambda \rangle = \lambda(\mu).$$

So e.g. $\langle \beta, \alpha^\checkmark \rangle = \alpha^\checkmark(\beta)$. Each $\alpha \in \Phi$ is called a root.

Example. If $\mathfrak{g}$ is a semisimple Lie algebra, and $\mathfrak{t} \subseteq \mathfrak{g}$ is the Cartan subalgebra, and Φ is the set of roots associated to $\mathfrak{t}$, then Φ is a root system in $\mathbb{R}\Phi$.

Definition 11.3 (Rank and isomorphism of root systems)**.**

1. The **rank** of a root system is $\dim_{\mathbb{R}}(E)$.
2. If (Φ, E), (Φ', E') are root systems, then an **isomorphism** of root systems is a linear isomorphism of spaces

   $$\rho : E \to E'$$

   with $\rho(\Phi) = \Phi'$, and

   $$\langle \rho(\alpha), \rho(\beta)^\checkmark \rangle = \langle \alpha, \beta^\checkmark \rangle \ \ \forall \alpha, \beta \in \Phi$$

Examples.

- **Rank 1:** A_1, $E = \mathbb{R}$, $(x, y) = xy$ gives the inner product. Then $\Phi = \{\pm\alpha\}$ where α is the unit vector
  $$\begin{pmatrix} 1 \\ 0 \end{pmatrix}$$
  $\langle \alpha, \alpha^\checkmark \rangle = 2$. ($w_\alpha$ is then refection in e.g. y-axis.)
- **Rank 2:** There are 4 cases here:
  1. $A_1 \times A_1$, $E = \mathbb{R}^2$, usual inner product. $\Phi = \{\pm\alpha, \pm\beta\}$. Standard basis vector $\pm e_1, \pm e_2$. Then w_α reflects α in the "y-axis" and w_β reflects β in the "x-axis".
  2. A_2, $\Phi = \{\pm\alpha, \pm\beta, \pm(\alpha + \beta)\}$. This is the root system for $\mathfrak{sl}_3$. (It looks like the vertices of a hexagon around the origin.)
  3. B_2, $\Phi = \{\pm\alpha, \pm\beta, \pm(\alpha + \beta), \pm(2\alpha + \beta)\}$. This is a root system for $\mathfrak{sp}_4$ and $\mathfrak{so}_5$. (It looks like the vertices of a square around the origin)
  4. G_2: See notes. This appears as a root system for the octonions. (It looks like the intersections of a hexagram centered at the origin.)

Returning to Lie algebras,

1. $\mathfrak{sl}_2 = \langle h \rangle \oplus \mathfrak{g}_\alpha \oplus \mathfrak{g}_{-\alpha}$ with $\alpha(h) = 2$. Taking h as the generator of the Cartan Subalgebra, we have root system corresponding to A_1.
2. $\mathfrak{sl}_3$ with
$$\mathfrak{t} = \left\langle \begin{pmatrix} 1 & & \\ & -1 & \\ & & 0 \end{pmatrix}, \begin{pmatrix} 0 & & \\ & 1 & \\ & & -1 \end{pmatrix} \right\rangle$$
has $\Phi = \{\pm\alpha, \pm\beta, \pm(\alpha+\beta)\}$. where $\alpha(h_1) = 2, \alpha(h_2) = -1$, $\beta(h_1) = -1, \beta(h_2) = 2$. Then
$$\Phi \cong A_2.$$
3. $\mathfrak{sp}_4$ or $\mathfrak{so}_5$, $\Phi \cong B_2$.
4. G_2 corresponds to a Lie algebra we haven't defined, which are the octionions $\mathbb{O}$.

Definition 11.4 (Weyl Group)**.** The **Weyl Group** of (Φ, E), an abstract root system, is the subgroup W of $\mathrm{GL}(E)$ generated by $\{w_\alpha : \alpha \in \Phi\}$. Where, recall
$$w_\alpha(\lambda) = \lambda - \frac{2(\alpha,\lambda)}{(\alpha,\alpha)}\alpha$$
sends α to $-\alpha$ and fixes everything on H_α

Exercise. W is finite: the w_α are invertible, and $w_\alpha\Phi \subseteq \Phi$. Each w_α permutes the elements of Φ, which is finite. Say $n = |\Phi|$, then there's an embedding
$$W \hookrightarrow \mathrm{Sym}(\Phi) \cong S_n$$
This map is an injection since $\mathbb{R}\Phi = E$. Therefore w_α, w_β act equally on Φ means that they agree on the whole of E.

Examples.

1. $A_1 : W \cong C_2$
2. $A_2 : W \cong D_6 (\cong S_3)$
3. $B_2 : W \cong D_8$
4. $G_2 : W \cong D_{12}$
5. $A_1 \times A_1 : W \cong V_4 \cong C_2 \times C_2$.

Lemma 11.5 (Reducible root systems)**.** *If (Φ_1, E_1), (Φ_2, E_2) are root systems, then*
$$(\Phi_1 \cup \Phi_2, E_1 \oplus E_2)$$

is also a root system. A root system of this form, with both $\Phi_i \neq \emptyset$ is called ***reducible****, otherwise it is* ***irreducible****.*

Examples.

1. $A_1 \times A_1$ is reducible
2. A_1, A_2, B_2, G_2 are all irreducible
3. If Φ corresponds to a Cartan subalgebra in a semisimple Lie algebra, then Φ is irreducible when $\mathfrak{g}$ is indecomposable. This is left as an exercise (but the solution is in his notes).

Lemma 11.6. *If Φ is a root system, and $\alpha, \beta \in \Phi$, with $\alpha \neq \pm\beta$. Then*

$$\langle \alpha, \beta^\vee \rangle \cdot \langle \beta, \alpha^\vee \rangle \in \{0, 1, 2, 3\}$$

Proof. $(\alpha, \beta) = \sqrt{(\alpha, \alpha)}\sqrt{(\beta, \beta)}\cos(\theta)$, where θ is the angle between α, β. So

$$\langle \alpha, \beta^\vee \rangle \cdot \langle \beta, \alpha^\vee \rangle = \frac{4(\alpha, \beta)^2}{(\alpha, \alpha)(\beta, \beta)} = 4\cos^2(\theta) \in \mathbb{Z}.$$

Now $\cos^2(\theta) \in [0, 1]$, therefore

$$\cos^2(\theta) \in \{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1\}$$

Since $\alpha \neq \pm\beta$, $\cos^2(\theta) \neq 1$. □

Corollary 11.7. *If Φ is a root system, and α,β are roots, then*

$$\langle \alpha, \beta^\vee \rangle \in \{0, \pm 1, \pm 2, \pm 3\}.$$

Exercise: Show that the only rank 2 systems are, up to isomorphism, those listed above. (Proof in the notes, non-trivial).

Corollary 11.8. *If Φ is an irreducible root system, then (α, α) can take at most two values as α varies over Φ.*

Proof. Left as an exercise (assume that there are 3 values, and look at ratios, and end up with something that is divisible by 6).

Definition 11.9 (Simply laced root system)**.** A root system Φ is called **simply laced** if all the roots have the same length (i.e. (α, α) takes one value for all α).

Examples. $A_1, A_1 \times A_1, A_2$ all are, but B_2 and G_2 are not.

Exercise: If Φ is simply laced, then (Φ, E) is isomorphic to a root system (Φ', E') with $\langle \alpha, \beta^\vee \rangle \in \{0, \pm 1\}$ for all α, β with $\alpha \neq \pm\beta$.

12 Weyl chambers and root bases

Throughout, (Φ, E) is a root system.

For $\alpha \in \Phi$,

$$H_\alpha = \{\lambda \in E : \langle \lambda, \alpha^\vee \rangle = 0\}$$

is the corresponding (root) hyperplane.

Definition (Weyl chambers)**.** The connected components of

$$E \setminus \bigcup_{\alpha \in \Phi} H_\alpha$$

are called **Weyl chambers**.

Definition (Root basis)**.** A subset

$$\Delta = \{\alpha_1, \ldots, \alpha_n\} \subseteq \Phi$$

is called a **root basis** or **base** if

1. Δ forms a basis for E.
2. $\alpha = \sum_{i=1}^{l} c_i \alpha_i \in \Phi \Rightarrow c_i \in \mathbb{Z}_{\geqslant 0}$ for all i, or $c_i \in \mathbb{Z}_{\leqslant 0}$ for all i.

We need to show that such a thing exists. First we define:

Definition 12.1 (Simple roots)**.** If $\Delta = \{\alpha_1, \ldots, \alpha_n\}$ is a base, then α_j are called **simple roots**. If $\alpha = \sum c_i \alpha_i$ with $c_i \geqslant 0$, then α is a **positive root** ($\alpha > 0$). If $c_i \leqslant 0$, then α is a **negative root** ($\alpha < 0$).

We then write

$$\Phi^+ = \text{ set of positive roots,}$$

and

$$\Phi^- = \text{ set of negative roots.}$$

it is clear that $\Phi^- = -\Phi^+$.

Remark. Δ defines a partial order on E.

$$\mu < \lambda \Leftrightarrow \lambda - \mu \text{ is a sum of positive roots, or } \lambda = \mu$$

Lemma 12.2. *Let W be the Weyl group of (Φ, E). Then if Δ is a base, $w \in W$, then $w(\Delta)$ is a base.*

Proof. We know that $w \in \mathrm{GL}(E)$, therefore $w(\Delta)$ is a basis for E, and a subset of Φ. If $\alpha \in \Phi$, and

$$\alpha = \sum c_i \alpha_i \text{ for } c_i \geqslant 0 \text{ or } c_i \leqslant 0 \text{ for all } i.$$

So

$$w(\alpha) = \sum c_i w(\alpha_i) \text{ for } c_i \geqslant 0 \text{ or } c_i \leqslant 0 \text{ for all } i.$$

□

Conclusion: W acts on the set of root bases.

The question remains: How do we construct them?

- Choose $\gamma \in E \setminus \cup_\alpha H_\alpha$, and define

  $$\Phi^+_\gamma = \{\alpha \in \Phi \,:\, \langle \gamma, \alpha^\vee \rangle > 0\}.$$

  Set $\Phi^-_\gamma = -\Phi^+_\gamma$.
- Define $\Delta_\gamma = \{\alpha \in \Phi^+_\gamma \,:\, \alpha \neq \beta_1 + \beta_2 \text{ for any } \beta_1, \beta_2 \in \Phi^+_\gamma\}$.

Theorem 12.3.

1. *Δ_γ is a root basis.*
2. *Every root basis is of the form Δ_γ for some $\gamma \in E \setminus \cup_\alpha H_\alpha$.*

Proof.

1. **Claim 1:** If $\alpha, \beta \in \Delta_\gamma$, $\alpha - \beta \notin \Delta_\gamma$.

   Suppose $\alpha, \beta \in \Phi$, WLOG $\alpha - \beta \in \Phi^+_\gamma$, (otherwise take $\beta - \alpha$). Then $\alpha = (\alpha - \beta) + \beta$ contradicting the definition of Δ_γ.

Claim 2: If $\alpha, \beta \in \Delta_\gamma$ and $\alpha \neq \beta$. Then $\langle \alpha, \beta \rangle = 0$. By (11.6), we know that $\langle \alpha, \beta^\vee \rangle . \langle \beta, \alpha^\vee \rangle \in \{0, 1, 2, 3\}$. Assume that $\langle \alpha, \beta^\vee \rangle > 0$. Without loss of generality, let it be 1 (otherwise, consider $\langle \beta, \alpha^\vee \rangle$). Now, we know that

$$\begin{aligned} w_\beta(\alpha) &= \alpha - \langle \alpha, \beta^\vee \rangle \beta \\ &= \alpha - \beta \in \Phi \end{aligned}$$

since W preserves Φ, which contradicts claim 1.

Claim 3: Let $\Delta_\gamma = \{\alpha_1, \ldots, \alpha_l\}$. If $\alpha \in \Phi_\gamma^+$, $\alpha = \sum_{i=1}^l c_i \alpha_i$, then $c_i \in \mathbb{Z}_{\geqslant 0}$. Suppose that there were an α which could not be witten in this way, pick such an α with (γ, α) minimal. Now $\alpha \notin \Delta_\gamma$, so $\alpha = \beta_1 + \beta_2$ for $\beta_1, \beta_2 \in \Phi_\gamma$. So

$$(\alpha\gamma) = (\beta_1, \gamma) + (\beta_2, \gamma),$$

and therefore $(\beta_1, \gamma) < (\alpha, \gamma)$. Both β_1 and β_2 can be written as a $\mathbb{Z}_{\geqslant 0}$-linear combination of the α_i, hence, as α is the sum of β_1 and β_2, so too can α. This gives us a contradiction.

Note that this means that every element in Φ_γ^- is a $\mathbb{Z}_{\leqslant 0}$ combination of α_i and that Δ_γ spans E, since Φ does.

Claim 4: Δ_γ forms a linearly independent set. Suppose that, for some $c_i \in \mathbb{R}$, $\sum_{i=1}^l c_i \alpha_i = 0$ Then without loss of generality, $c_i \geqslant 0$ for $1 \leqslant i \leqslant m$, and $c_i \leqslant 0$ for $m+1 \leqslant i \leqslant l$. So

$$v := \sum_{i=1}^m c_i \alpha_i = -\sum_{j=m+1}^l c_j \alpha_j$$

And therefore

$$0 \leqslant (v, v) = -\sum_{i,j} c_i c_j (\alpha_i, \alpha_j) \leqslant 0.$$

Since $c_i c_j \leqslant 0$, and by claim 2, $(\alpha_i, \alpha_j) \leqslant 0$, so $v = 0$. Hence $0 = (\gamma, v) = \sum_{i=1}^m c_i (\gamma, \alpha_i)$. So $c_i = 0$ for $1 \leqslant i \leqslant m$, and similarly for $m+1 \leqslant j \leqslant l$.

2. We provide a sketch of the proof from Humphreys, section 10.

   Given a base Δ, select $\gamma \in E$ such that $(\gamma, \alpha) > 0)$ for all $\alpha \in \Delta$, this is possible, since the intersection of "positive" open half spaces associated with any basis of E is nonempty. Now $\gamma \in E - \cup_\Phi H_\alpha$ and $\Phi^+ \subseteq \Phi_\gamma^+$, $\Phi^- \subseteq -\Phi_\gamma^+$. So equality must hold in both cases. Since $\Phi^+ = \Phi_\gamma^+$, Δ clearly consists of indecomposable[13] elements, i.e. $\Delta \subseteq \Delta_\gamma$. But $|\Delta| = |\Delta_\gamma| = l$, therefore $\Delta = \Delta_\gamma$. □

[13] where $\alpha \in \Phi_\gamma^+$ is decomposable if it can be written $\alpha = \beta_1 + \beta_2$, where $\beta_1, \beta_2 \in \Phi_\gamma^+$, indecomposable otherwise.

Corollary 12.4. *There is a bijection*

$$\{\textit{Weyl Chambers}\} \leftrightarrow \{\textit{root bases}\}.$$

Proof. Clear, given a Weyl chamber C, choose $\gamma \in C$ and choose a base Δ_γ. Given $\Delta = \Delta_\gamma$, take $\gamma \in C$ for some Weyl chamber C. □

Definition 12.5 (Height)**.** If $\Delta = \{\alpha_1, \ldots, \alpha_l\}$ is a base and $\alpha \in \Phi$, then the **height** of α is

$$\sum_{i=1}^{l} c_i,$$

where $\alpha = \sum_{i=1}^{l} c_i \alpha_i$.

Lemma 12.6. *If $\Delta = \{\alpha_1, \ldots, \alpha_l\}$ is a base, and $\beta \in \Phi^+ \setminus \Delta$, then there exists i such that $\beta - \alpha_i \in \Phi$.*

Proof. Given β, if $(\beta, \alpha_i) \leqslant 0$ for all i, then $\Delta \cup \{\beta\}$ is a linearly independent set by the proof of claim 4 above. So there must exist an i such that $\langle \beta, \alpha_i^\checkmark \rangle > 0$. Since $\langle \beta, \alpha_i^\checkmark \rangle \langle \alpha_i, \beta^\checkmark \rangle \in \{0, 1, 2, 3\}$, then we must have $\langle \beta, \alpha_i^\checkmark \rangle = 1$ or $\langle \alpha_i, \beta^\checkmark \rangle = 1$. So $w_{\alpha_i}(\beta) = \beta - \alpha_i \in \Phi$, or $w_\beta(\alpha_i) = \alpha_i - \beta \in \Phi$. Therefore $\beta - \alpha_i \in \Phi$. □

Corollary 12.7. *If $\beta \in \Phi^+$, then β can be written as $\beta = \sum_{j=1}^{n} \alpha_{ij}$ with α_{ij} a simple root for each j, and wherew $\sum_{j=1}^{k} \alpha_{ij}$ is a root for all k.*

Proof. This is clear by induction on the height, using (12.6) □

This implies that for $\mathfrak{g}$ a semisimple Lie algebra, with cartan subalgebra $\mathfrak{t}$, and root system Φ, then given a root basis Δ, $\mathfrak{g}$ is generated by

$$\{e_\alpha, e_{-\alpha} : \alpha \in \Delta\}.$$

13 Facts about the Weyl group

Recall that W acts on the set of root bases, and so preserves the set of Weyl chambers. Using the following lemma

Lemma 13.1. *If $w \in W$ and $\lambda, \mu \in E$, then $\langle \lambda, \mu^\vee \rangle = \langle w(\lambda), w(\mu)^\vee \rangle$.*

We can deduce the following proposition:

Proposition 13.2. *If Δ is a root basis and $w \in W$, then*

$$C_w(\Delta) = w(C_\Delta).$$

Lemma 13.3. *For Φ a root system, Δ a root basis, and W a Weyl group, $\alpha \in \Delta$, w_α permutes $\Phi^+ \setminus \Delta$.*

Proof. Take $\alpha = \alpha_1$, $\Delta = \{\alpha_1, \ldots, \alpha_l\}$. Suppose $\beta \in \Phi^+ \setminus \{\alpha\}$. Then $\beta = \sum c_i \alpha_i$ with $c_i \in \mathbb{Z}_{\geqslant 0}$.

$$w_{\alpha_1}(\beta) = \beta - \langle \beta, \alpha_1^\vee \rangle \alpha_1 = (c_1 - \langle \beta, \alpha_1^\vee \rangle)\alpha_1 + \sum_{i=2}^{l} c_i \alpha_i.$$

Since β is a positive root, and $\beta \neq \alpha_1$, $w_{\alpha_1}(\beta) \neq -\alpha_1$, we must have $c_i > 0$ for some $i \geqslant 2$, hence $w_{\alpha_1}(\beta)$ is a positive root *different from* α, since $\beta \neq \alpha$. □

Theorem 13.4.
1. *The Weyl group acts simply transitively (action is transitive, and stabiliser of any point is trivial) on the set of root bases (and on the set of Weyl chambers).*
2. *Given a root basis Δ and $\alpha \in \Phi$, there's an element $w \in W$ with $w(\alpha) \in \Delta$. This w is not necessarily unique.*
3. *If $\Delta = \{\alpha_1, \ldots, \alpha_l\}$ is a root basis, then W is generated by $\{w_{\alpha_i} \mid 1 \leqslant i \leqslant l\}$*

Proof. In Humphreys' book. Omitted.

14 Classification of irreducible root systems

Throughout (Φ, E) is a root system, $\Delta = \{\alpha_1, \ldots, \alpha_l\}$ is a root basis, and W is the Weyl group of Φ. Then we define:

Definition (Cartan Matrix)**.** The **Cartan matrix** of Φ is the $l \times l$ matrix

$$C = (a_{ij})_{1 \leqslant i,j \leqslant l}$$

with

$$a_{ij} = \langle \alpha_i, \alpha_j^{\checkmark} \rangle$$

This is independent of the choice of root basis, since given Δ'. $\exists w \in W$ with $w(\Delta) = \Delta'$. This action preserves the angle brackets, and hence the Cartan matrix. Note that $\det C \neq 0$ since C is obtained by taking inner products of a basis.

Example. For G_2 if we let $\alpha_1 = \alpha$ and $\alpha_2 = \beta$, then

$$\langle \alpha_1, \alpha_2^{\checkmark} \rangle = -1$$
$$\langle \alpha_2, \alpha_1^{\checkmark} \rangle = -3$$

So we obtain cartan matrix

$$\begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix}$$

with uniqueness up to reordering the base. Now let's quickly go through some more:

For $A_1 \times A_1$:

$$\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$$

For A_2:

$$\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$$

For B_2:

$$\begin{pmatrix} 2 & -2 \\ -1 & 2 \end{pmatrix}$$

Proposition 14.1. *Suppose (Φ', E') is a root system, with base $\{\alpha_1', \ldots, \alpha_l'\}$ such that*

$$\langle \alpha_i, \alpha_j^{\checkmark} \rangle = \langle \alpha_i', (\alpha_j')^{\checkmark} \rangle \ \ \forall i, j$$

Then the linear map

$$\alpha_i \mapsto \alpha_i'$$

induces an isomorphism Φ of root systems such that

$$\langle \phi(\alpha), \phi(\beta)^{\checkmark} \rangle = \langle \alpha, \beta^{\checkmark} \rangle$$

for all $\alpha, \beta \in \Phi$. Thus the Cartan matrix of Φ determines Φ up to isomorphism.

Proof. Since Δ (respectively Δ') is a basis of E (respectively E'), there exists a unique vector space isomorphism

$$\begin{aligned} \phi : E &\to E' \\ \alpha_i &\mapsto \alpha_i'. \end{aligned}$$

If $\alpha, \beta \in \Delta$, then

$$\begin{aligned} w_{\phi(\alpha)}(\phi(\beta)) &= w_{\alpha'}(\beta') \\ &= \beta' - \langle \beta', (\alpha')^\vee \rangle \alpha' \\ &= \phi(\beta) - \langle \beta, \alpha^\vee \rangle \phi(\alpha) \\ &= \phi(\beta - \langle \beta, \alpha^\vee \rangle \alpha) \\ &= \phi(w_\alpha(\beta)) \end{aligned}$$

That is to say, we have a commutative diagram

$$\begin{array}{ccc} E & \xrightarrow{\phi} & E' \\ \downarrow{\scriptstyle w_\alpha} & & \downarrow{\scriptstyle w_{\phi(\alpha)}} \\ E & \xrightarrow{\phi} & E' \end{array}$$

Now use (13.3). THe respective Weyl groups W and W' are generated by simple reflections, therefore the map

$$w \mapsto \phi \circ w \circ \phi^{-1}$$

is an isomorphism $W \to W'$ sending $w_\alpha \to w_{\phi(\alpha)}$ for $\alpha \in \Delta$. Each $\beta \in \Phi$ is conjugate under W to a simple root, say $\beta = w(\alpha)$ for $\alpha \in \Delta$. This implies

$$\phi(\beta) = (\phi \circ w \circ \phi^{-1})(\phi(\alpha)) \in \Phi'$$

Hence ϕ maps Φ onto Φ'. The formula for reflections shows that ϕ preserves cartan integers, i.e.:

$$\langle \alpha, \beta^\vee \rangle = \langle w(\phi(\alpha)), w(\phi(\beta))^\vee \rangle.$$

□

Remark. The proposition suggests that it is possible in principle to recover Φ from a knowledge of the Cartan integers, see Humphreys' book, page 56 for the algorithm for doing this.

Definition 14.2 (Coxeter Graph)**.** Recall that if $\alpha \neq \pm\beta$, then

$$\langle \alpha, \beta^\vee \rangle \langle \beta, \alpha^\vee \rangle \in \{0, 1, 2, 3\}$$

The **Coxeter graph** of Φ is a graph with l vertices, where the ith vertex is joined to the jth by $\langle \alpha_i, \alpha_j^\vee \rangle \langle \alpha_j, \alpha_i^\vee \rangle$ edges. The graph determines $\langle \alpha_i, \alpha_j^\vee \rangle$ in the case that all roots have equal length.

In the case that more than one root length occurs (e.g. B_2, G_2) the graph fails to tell us which of a pair of vertices should correspond to a short simple root, and which to a long simple root (in the case where the vertices are joined by 2 or 3 edges).

Remark. The Coxeter graph does determine the Weyl group W completely, as it determines the orders of products of generators of W.

Definition (Dynkin Diagram)**.** The **Dynkin diagram** of Φ has

- Vertices $\leftrightarrow \Delta$
- i^{th} and j^{th} vertices are connected by $\langle \alpha_i, \alpha_j^\vee \rangle \langle \alpha_j, \alpha_i^\vee \rangle$ edges.
- If a multiple edge occurs then an arrow points to which one is the shorter root (to avoid the indeterminacy).

Examples 14.3. For ranks 1 and 2, we have the following.

- Type A_1: $\bullet$.
- Type $A_1 \times A_1$: $\bullet \ \bullet$
- Type A_2: $\bullet\!\!-\!\!\bullet$
- Type B_2: $\bullet\!\!\Rightarrow\!\!\bullet$
- Type G_2: $\bullet\!\!\Lleftarrow\!\!\bullet$

Lecture 10

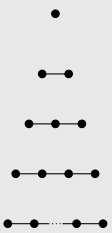
Remark. The maximal number of edges between any pair of vertices is 3, and Φ is simply laced if and only if the Dynkin diagram has no multiple edges.

Exercise: Φ is irreducible if and only if the Dynkin diagram is simply connected.

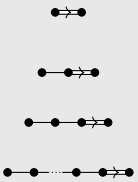
Theorem 14.4. *If Φ is an irreducible root system (of rank l in the first batch) then its Dynkin diagram is one of either*

- *A diagram associated to the classical root systems:*

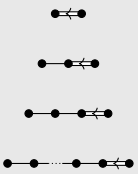
- *A_l where $l \geqslant 1$ is an open polygon (i.e. a straight line):*



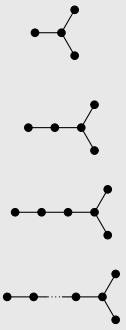
- *B_l where $l \geqslant 2$ is the same, but with a double arrow at the end:*



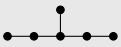
- *C_l where $l \geqslant 2$ is the same as B_l with the arrow pointing the other way:*



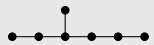
- *D_l where $l \geqslant 4$ has the diagram bifurcating at the end:*



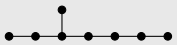
- *Or it is one of the following exceptional root systems*
  - *E_6:*



– E_7:



– E_8:



– F_4:

– G_2:

Proof. Omitted, can be found [Humphreys, 11.4]. It takes him 5 pages. Worth reading at least once in your life when you're really bored. □

Theorem 14.5. *For every Dynkin diagram $\mathcal{D}$ listed, there is a simple Lie algebra $\mathfrak{g}$ with Cartan subalgebra $\mathfrak{t}$, and roots corresponding to $\mathfrak{t}$ such that the Dynkin diagram of this root system, Φ, is $\mathcal{D}$.*

Proof. Sketch here, full proof found in Humphreys.

For A_l, let e_i be the ith standard basis vector in an $l+1$ dimensional Euclidean space, $\mathbb{R}^{l+1}$. Let

$$\Phi = \{e_i - e_{j\,:\,i\neq j} \subseteq \mathbb{R}^{l+1}.$$

Then Φ spans an l-dimensional subspace of $\mathbb{R}^{l+1}$: call this E. Then Φ is a root system in E with root basis given by:

$$\{\alpha_i = e_i - e_{i+1} \,|\, 1 \leqslant i \leqslant l\}.$$

(Notice that $e_i - e_j = (e_i - e_{i+1}) + (e_{i+1} - e_{i+2}) + \cdots + (e_{j-1} - e_j)$ if $i < j$).

Also note that

$$\langle \alpha_i, \alpha_j^{\checkmark} \rangle = \begin{cases} 1, & i, j \text{ differ by 1} \\ 2, & i = j \\ 0, & \text{otherwise} \end{cases}$$

The corresponding Dynkin diagram is

(i.e. A_l) since we have α_1 connected with one edge to α_2 connected with ... to α_l. Now w_{α_i} flips the ith and $(i+1)^{\text{th}}$ coordinates, so $w \cong S_{l+1}$. The corresponding Lie algebra is $\mathfrak{sl}_{l+1}(\mathbb{C})$ with Cartan subalgebra given by

$$\mathfrak{t} = \{\text{ diagonal matrices }\}$$

and

$$\alpha_i\left(\begin{pmatrix} t_1 & & \\ & \ddots & \\ & & t_{l+1} \end{pmatrix}\right) = t_i - t_{i+1}$$

With Cartan matrix

$$\begin{pmatrix} 2 & -1 & 0 & \cdots & 0 \\ -1 & 2 & -1 & & \\ 0 & -1 & 2 & & \\ \vdots & & & \ddots & \\ 0 & & & & 2 \end{pmatrix}.$$

For the other classical root systems, with e_i a basis of $\mathbb{R}^l$, and $\mathfrak{t}$ the diagonal subalgebra, we get

Type	$\Phi \subseteq \mathbb{R}^{l+1}$	$\Delta \subseteq \Phi$	W	$\mathfrak{g}$	dim
B_l	$\{\pm e_i, \pm e_i \pm e_j \,:\, i \neq j\}$	$\{e_i - e_{i+1} \,:\, 1 \leqslant i \leqslant l\} \cup \{e_l\}$	$S_l \ltimes C_2^l$	$\mathfrak{so}_{2l+1}$	$2l^2 + l$
C_l	$\{\pm 2e_i, \pm e_i \pm e_j \,:\, i \neq j\}$	$\{e_i - e_{i+1} \,:\, 1 \leqslant i \leqslant l\} \cup \{2e_l\}$	$S_l \ltimes c_2^l$	$\mathfrak{sp}_{2l}$	$2l^2 + l$
D_l	$\{\pm e_i \pm e_j \,:\, i \neq j\}$	$\{e_i - e_{i+1} \,:\, 1 \leqslant i \leqslant l\} \cup \{e_{l-1} + e_l\}$	$S_l \ltimes C_2^{l-1}$	$\mathfrak{so}_{2l}$	$2l^2 - l$

For the exceptional types, we have:

- G_2: 12 roots,
$$E = \{w = \sum_{i=1}^{3} c_i e_i \in \mathbb{R}^3 \,:\, \sum c_i = 0\}$$
done in Humphreys, page 65.
- F_4: $E = \mathbb{R}^4$, $|\Phi| = 48$, $|W| = 2^7 3^2$.
- For E_6, E_7, E_8: first do E_8 then find root systems of types E_7, E_6 inside it. For E_8, $|\Phi| = 240$, $|W| = 2^{14}3^5 5^2 7$.

Let $w_c = \prod_1^8 w_{\alpha_i}$, a Coxeter element of W of order 30. There is a plane of $\mathbb{R}^8$ on which this coxeter element acts by rotation.

Remark.

1. To look up root systems, use the "spherical explorer" website. Link here.
2. When doing computations, always helpful to compute things in terms of simple roots.
3. The exceptional Lie algebras are written as:

$$\mathfrak{g}_2 \qquad \mathfrak{f}_4 \qquad \mathfrak{e}_6 \qquad \mathfrak{e}_7 \qquad \mathfrak{e}_8$$

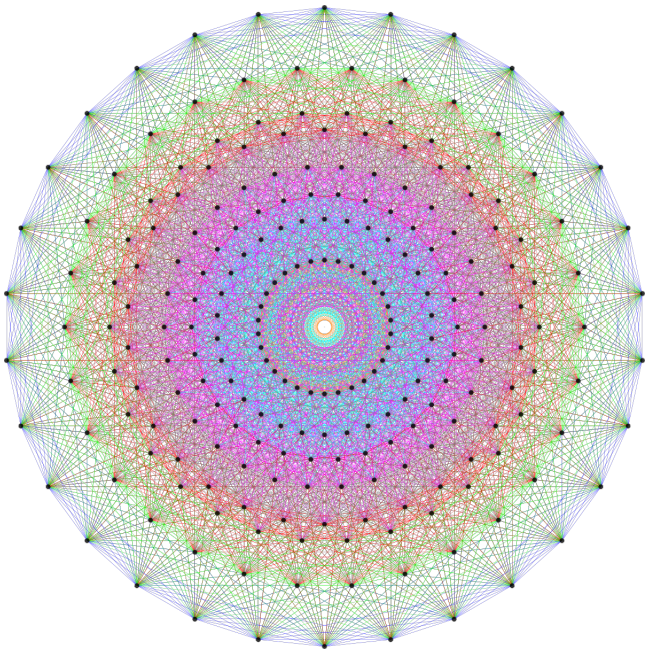


Figure 1: This "is" the root lattice of E_8. By that, we mean it is a projection into 2-dimensions of the 8 dimensional root lattice system. I'm not too sure that this picture is the right way to approach analysing E_8, but it's pretty.

$\mathfrak{g}_2$ is the algebra of "derivations of the octonions" $\mathbb{O}$, where a **derivation**[14] is a linear map δ such that

$$\delta(ab) = \delta(a)b + a\delta(b)$$

(where multiplication is taken to be a bracket in the Lie algebra case). $\mathbb{O}$ is an 8-dimensional normal division algebra over $\mathbb{R}$. It has a 1 dimensional centre on which $\mathfrak{g}_2$ acts trivially. Moreover, there is a representation

$$\mathfrak{g}_2 \to \mathfrak{so}_7$$

which is the lowest dimensional non-trivial representation of $\mathfrak{g}_2$ (More can be found in [Humphreys, 19.3]). Other can be constructed ([Fulton and Harris, 22.4]).

4. Given Φ, there is a natural construction of a Lie algebra with that root system.

Lecture 11 We've seen that there is a correspondence

[14] On Example Sheet 2, for reference

$$\{\,\mathfrak{g} \text{ simple, } \mathfrak{t} \text{ CSA}\} \longrightarrow\!\!\!\!\!\rightarrow \{\,\text{irred root systems } \Phi\,\}$$
$$\updownarrow$$
$$\{\,\text{connected Dynkin diagrams}\,\}$$

Lecture 12 Now we want to:

- Show that the root system corresponding to $\mathfrak{g}$ is independent of the choice of CSA.
- Show that two Lie algebras with the same root system are isomorphic.

9 Brief introduction to inner automorphisms

Remember section 9? Here it is!

Definition (Automorphism)**.** An **automorphism** of $\mathfrak{g}$ is an isomorphism $\mathfrak{g} \to \mathfrak{g}$. The gorup of all such is $\operatorname{Aut}\mathfrak{g}$. For example, if $\mathfrak{g} = \mathfrak{gl}(V)$ or $\mathfrak{sl}(V)$ and $A \in \mathrm{GL}(V)$ is an invertible endomorphism $V \to V$, then

$$x \mapsto AxA^{-1}$$

is an automorphism of $\mathfrak{g}$.

Take V finite dimensional over $\mathbb{C}$. Take $x \in \mathfrak{g}$ such that $\operatorname{ad} x$ is nilpotent (i.e. $(\operatorname{ad} x)^m = 0$ for some $m > 0$). Then

$$\exp(\operatorname{ad} x) = 1 + \operatorname{ad} x + \frac{(\operatorname{ad} x)^2}{2!} + \cdots + \frac{(\operatorname{ad} x)^{m-1}}{(m-1)!}$$

makes sense to define, as there are finitely many terms. It is easy to show that $\exp(\operatorname{ad} x) \in \operatorname{Aut}(\mathfrak{g})$. Any automorphism of the form $\exp(\operatorname{ad} x)$, where $\operatorname{ad} x$ is nilpotent, is called **inner**. The subgroup of $\operatorname{Aut}\mathfrak{g}$ generated by these is denoted $\operatorname{Inn}\mathfrak{g}$. We can see that it's normal: if $\phi \in \operatorname{Aut}\mathfrak{g}$, $x \in \mathfrak{g}$, then

$$\phi(\operatorname{ad} x)\phi^{-1} = \operatorname{ad}\phi(x).$$

so we must have

$$\phi \exp(\operatorname{ad} x)\phi^{-1} = \exp(\operatorname{ad}\phi(x))$$

Lemma 9.1. *If $\mathfrak{g} \leqslant \mathfrak{gl}(V)$ (over $\mathbb{C}$) and x is nilpotent, then $\operatorname{ad} x$ is nilpotent and*

$$\exp(x)y(\exp x)^{-1} = \exp(\operatorname{ad} x)(y) \quad \forall y \in \mathfrak{g}.$$

Proof. Humphreys, page 9 (I think). □

Examples.

1. $\mathfrak{g} = \mathfrak{sl}_n(\mathbb{C})$: $\mathrm{Inn}(\mathfrak{g}) = \mathrm{PGL}_n(\mathbb{C})$.
2. $\mathfrak{g} = \mathfrak{so}_n(\mathbb{C})$: $\mathrm{Inn}(\mathfrak{g}) = \mathrm{SO}_n / Z$
3. $\mathfrak{g} = \mathfrak{sp}_{2n}(\mathbb{C})$: $\mathrm{Inn}(\mathfrak{g}) = \mathrm{Sp}_{2n}(\mathbb{C})/Z$

15 Conjugacy results

Here $\mathfrak{g}$ is a semisimple Lie algebra, $\mathfrak{t}$ is a CSA of $\mathfrak{g}$, Φ is a root system corresponding to $\mathfrak{t}$, i.e.:

$$\mathfrak{g} = \mathfrak{t} \oplus \bigoplus_{\alpha \in \Phi} \mathfrak{g}_\alpha,$$

and $\Delta \subseteq \Phi$ is a root basis.

Lemma 15.1. *If $\mathfrak{t}'$ is another CSA of $\mathfrak{g}$ then there is an inner automorphism $\Psi \in \mathrm{Inn}(\mathfrak{g})$ with $\Psi(\mathfrak{t}) = \mathfrak{t}'$.*

Proof. [Humphreys 16.4] or [Carter, p34]. Pretty heavy-duty group theory. □

Definition (Rank)**.** Define the **rank** of $\mathfrak{g}$ as the dimension of a CSA which is independent of choice of CSA by (15.1). If $\mathfrak{g}$ is semisimple, then $\operatorname{rank}\mathfrak{g}$ coincides with $\operatorname{rank}\Phi$, where Φ is the root system of $\mathfrak{g}$ relative to any maximal toral subalgebra (i.e. relative to any CSA).

Lemma 15.2. *If $\mathfrak{t}'$ is a CSA of $\mathfrak{g}$ with root system Φ, then Φ and Φ' are isomorphic.*

Proof. Take $\Psi \in \operatorname{Inn}\mathfrak{g}$ as in (15.1), mapping $\mathfrak{t}$ to $\mathfrak{t}'$. Take $t \in \mathfrak{t}$, $\alpha \in \Phi$, and $e_\alpha \in \mathfrak{g}\alpha$. Then

$$[\Psi(t), \Psi(e_\alpha)] = \Psi([t, e_\alpha])$$

since Ψ is a homomorphism, so respects the bracket. Then

$$\begin{aligned} \Psi([t, e_\alpha]) &= \Psi(\alpha(t)e_\alpha) \\ &= \alpha(t)\Psi(e_\alpha) \end{aligned}$$

Then as $\Psi(e_\alpha)$ spans the root space for $\mathfrak{t}'$,

$$\Phi' = \{\alpha \circ \Psi^{-1} \,:\, \alpha \in \Phi\}.$$

□

Theorem 15.3. *Suppose $\mathfrak{g}'$ is a semisimple Lie algebra with root system Φ, then $\mathfrak{g} \cong \mathfrak{g}'$.*

Proof. In [Carter, Chapter 7] using the theory of finite structure constants. The idea is to choose a basis h_{α_i} of $\mathfrak{t}$, and choose e_α in each root space, so that for each α:

$$[e_\alpha, e_{-\alpha}] = h_\alpha.$$

This gives a basis of the Lie algebra. Then

$$[h_{\alpha_i}, h_{\alpha_j}] = 0, \quad [h_{\alpha_i}, e_\alpha] = \alpha(h_{\alpha_i})e_{\alpha_i}, \quad [e_\alpha, e_\beta] = \begin{cases} N_{\alpha\beta}e_{\alpha+\beta}, & \alpha+\beta \in \Phi \\ h_\alpha, & \beta = -\alpha \\ 0, & \alpha+\beta \notin \Phi \cup \{0\} \end{cases}$$

16 Weights

Example. Let $\mathfrak{g} = \mathfrak{so}_5(\mathbb{C})$, this is type B_2 (which is actually the same as type C_2[15]), α, β are simple roots for the root system of $\mathfrak{g}$, which we call Φ. Recall, we had $\mathfrak{m}_\alpha$ given by:

$$\mathfrak{m}_\alpha = \mathfrak{g}_\alpha \oplus \langle [\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}] \rangle \oplus \mathfrak{g}_{-\alpha} \cong \mathfrak{sl}_2.$$

Decompose the adjoint representation of $\mathfrak{g}$ under the actions of $\mathfrak{m}_\alpha$, $\mathfrak{m}_\beta$.

If $e_\alpha \in \mathfrak{g}_\alpha$, then $e_\alpha \mathfrak{g}_\gamma = \mathfrak{g}_{\alpha+\gamma}$ for all $\gamma \in \Phi$. So each α-root string corresponds to an irreducible subrepresentation ($\mathfrak{m}_\alpha$) of $\mathfrak{g}$.

$$\mathfrak{g}|_{\mathfrak{m}_\alpha} = V(2) \oplus V(2) \oplus V(2) \oplus V(0).$$

Similarly:

$$\mathfrak{g}|_{\mathfrak{m}_\beta} = V(2) \oplus V(1) \oplus V(1) \oplus V(0) \oplus V(0) \oplus V(0).$$

Now (Φ, E) is a root system, and fix the base $\Delta = \{\alpha_1, \ldots, \alpha_l\}$.

Definition 16.1 (Root lattice/Weight lattice)**.** The **root lattice** $\mathbb{Z}\Phi$ is defined as

$$\mathbb{Z}\Phi = \{\sum_{\alpha \in \Phi} c_\alpha \alpha, \ c_\alpha \in \mathbb{Z}\} \subseteq E.$$

The **weight lattice** X is defined:

$$X = \{\lambda \in E, \ \langle \lambda, \alpha^\vee \rangle \in \mathbb{Z}, \ \forall \alpha \in \Phi\}.$$

[15] Which we technically did not allow to exist when we defined these.

Or $\{\beta \in \mathfrak{t}^* \,:\, \beta(h_\alpha) \in \mathbb{Z} \ \forall \text{ roots } \alpha\}$.

The Weyl group acts on X

Note:

- $\mathbb{Z}\Phi \subseteq X$.
- If $\lambda \in X$ so is $w(\lambda)$ (since $\langle \lambda, \alpha^\vee \rangle = \langle w(\lambda), w(\alpha)^\vee \rangle$)).
- The root lattice *is* a lattice in E in the technical sense (i.e. the $\mathbb{Z}$-span of an $\mathbb{R}$-basis, i.e. it's a finitely-generated free abelian group, so isomorphic to some $\mathbb{Z}^l$, with form $(\,\cdot\,,\,\cdot\,) : L \times L \to \mathbb{Z}$, such that $(L \otimes_{\mathbb{Z}} \mathbb{R},\ (\,\cdot\,,\,\cdot\,))$ is an inner product space.

Lemma 16.2. *$\lambda \in X$ if and only if $\langle \lambda, \alpha^\vee \rangle \in \mathbb{Z}$, $\forall \alpha \in \Delta$.*

Definition 16.3 (Fundamental weights with respect to Δ)**.** For $1 \leqslant i \leqslant l$, define $\omega_i \in E$ by

$$\langle \omega_i, \alpha_j^\vee \rangle = \delta_{ij}$$

and call $\{\omega_i\}$ the **fundamental weights with respect to Δ**.

By (16.2),

$$X = \left\{ \sum_i c_i \omega_i \,:\, c_i \in \mathbb{Z} \right\}.$$

Exercise: $X/\mathbb{Z}\Phi$ is a finite group, called the **fundamental group**. In fact, the index $|X/\mathbb{Z}\Phi| = \det C$ where C is the Cartan matrix. [Humphreys, page 68].

Example. Consider $\mathfrak{g} = \mathfrak{sl}_2$, with type A_1, and roots $\{\alpha, -\alpha\}$. The root lattice $\mathbb{Z}\Phi$ is then just $\mathbb{Z}\alpha$, and $\langle \alpha, \alpha^\vee \rangle = 2$, and $X = \mathbb{Z}\frac{\alpha}{2}$. Here $|X/\mathbb{Z}\Phi| = 2 = \det(2) = \det(C)$. In fact, for A_l, $|X/\mathbb{Z}\Phi|$ is always $l+1$. For type B_l, it's always l, and for G_2, $X = \mathbb{Z}\Phi$, so the index[16] is 1.

Definition 16.4 (Dominant)**.** A $\lambda \in X$ is **dominant** if

$$\langle \lambda, \alpha^\vee \rangle \geqslant 0, \ \forall \alpha \in \Phi^+,$$

strongly dominant if > 0. X^+ is the set of dominant weights (sometimes called the cone of dominant weights). This is equivalent to:

[16] Which we call the "index of connection".

- λ lies in the closure of the fundamental Weyl chamber with respect to Δ (I assume this means the Weyl chamber that γ lies in, where $\Delta = \Delta_\gamma$).
- $\lambda = \sum_{i=1}^{l} c_i w_i$ with all $c_i \in \mathbb{Z}_{\geqslant 0}$. ($X^+ = \bigoplus_{i=1}^{l} \mathbb{Z}_{\geqslant 0}\omega_i$)).

Assume $\mathfrak{g}$ is semisimple, with root system Φ, Cartan subalgebra $\mathfrak{t}$. Choose $e_\alpha \in \mathfrak{g}_\alpha$ for each $\alpha \in \Phi$, such that $[e_\alpha, e_{-\alpha}] = h_\alpha$, and $\varphi : \mathfrak{g} \to \mathfrak{gl}(V)$ a finite dimension representation over $\mathbb{C}$.

Lemma 16.5 (Weight Space Decomposition)**.** $V = \bigoplus_{\lambda \in \mathfrak{t}^*} V_\lambda$ *where*

$$V_\lambda = \{v \in V \,:\, tv = \lambda(t)v,\ \forall t \in \mathfrak{t}\}$$

Proof. Clear from Lemma 10.2, where the commuting semisimple endomorphisms are the basis elements of $\mathfrak{t}$. □

Some Terminology:

Recall that for $\lambda, \mu \in \mathfrak{t}^*$, we write $\mu \leqslant \lambda$ to mean $\lambda - \mu = \sum c_i \alpha_i$ for $\alpha_i \in \Delta$, and $c_i \in \mathbb{Z}_{\geqslant 0}$.

And if V is a representation of $\mathfrak{g}$, we say:

- The **weight** of vector $0 \neq v \in V$ is λ if $v \in V_\lambda$.
- $\lambda \in \mathfrak{t}^*$ is a highest weight if $V_\lambda \neq 0$ and if $V_\mu \neq 0$ for all $\mu \leqslant \lambda$ (in the above ordering).

Proposition 16.6.

1. *If $v \in V_\lambda$ then $e_\alpha V_\lambda \in V_{\alpha+\lambda}$.*
2. *If $V_\lambda \neq 0$, then $\lambda \in X$ (i.e. $\lambda(h_\alpha) \in \mathbb{Z}$ for all α).*
3. $\dim V_\lambda = \dim V_{w\lambda}$.

Proof.

1. Fix $t \in \mathfrak{t}$. Then
$$\begin{aligned} te_\alpha v &= ([te_\alpha] + e_\alpha t)v \\ &= \alpha(t)e_\alpha(v) + e_\alpha \lambda(t)v \\ &= (\alpha + \lambda)(t)e_\alpha v \end{aligned}$$
2. Consider $V|_{\mathfrak{m}_\alpha}$. Then h_α acts by integer weights, so $\lambda(h_\alpha) \in \mathbb{Z}$.

3. Enough to assume $w = w_\alpha$ for some $\alpha \in \Phi$ (since they generate the Weyl group).

   Now $V|_{\mathfrak{m}_\alpha} = \bigoplus V^{(j)}$, where $V^{(j)}$ are $\mathfrak{m}_\alpha$-irreducible representations. Since h_α-weight spaces of $V^{(j)}$ are 1-dimensional, we can choose a basis $\{v_1, \dots, v_n\}$ for V_λ, with each v_i in a distinct $V^{(j)}$.

   Now it's enough to prove that given $v_i \in V^{(j)}$, $\exists x \in \mathfrak{m}_\alpha$ such that $xv_i \in V_{w_\alpha \lambda}$. We know $w_\alpha(\lambda) = \lambda - \langle \lambda, \alpha^\vee \rangle \alpha$, and that $\{e_{-\alpha}^k v_i, e_\alpha^k v_i \,:\, k \in \mathbb{Z}_{\geqslant 0}\}$ spans $V^{(j)}$ (this comes from $\mathfrak{sl}_2$-theory we developed before, i.e. jumping left and right).

   So define
   $$M = \max\{k \,:\, e_\alpha^k v_i \neq 0\}$$
   and
   $$m = \max\{k \,:\, e_{-\alpha}^k v_i \neq 0.$$

   We need to prove $-m \leqslant \langle \lambda, \alpha^\vee \rangle \leqslant M$ as in (10.13(2)). However:
   $$(\lambda + M_\alpha)(h_\alpha) = -(\lambda - m_\alpha)(h_\alpha)$$
   $$\therefore \lambda(h_\alpha) = m - M,$$
   and then since $\lambda(h_\alpha) = \langle \lambda, \alpha^\vee \rangle$, we're done.

□

Definition 16.7 (Highest weight vector/HWV)**.** $v \in V$ is a **highest weight vector** (HWV) if

- $v \neq 0$.
- $v \in V_\lambda$, for some λ.
- $e_\alpha(v) = 0$, $\forall \alpha \in \Phi^+$.

On the example sheet, we show that there is a root α_0 of maximal height with respect to Δ (the **highest root**). Any non-zero $v \in \mathfrak{g}_{\alpha_0}$ is a highest weight vector with respect to the adjoint representation.

Example. $\mathfrak{g} = \mathfrak{sl}_3$, $\mathfrak{t}$ is the CSA, the root lattice is a series of points (which looks like a lattice of equilateral triangles, picture it) in $\mathbb{R}^2$, and basis for the CSA:
$$h_{\alpha_1} = \begin{pmatrix} 1 & & \\ & -1 & \\ & & 0 \end{pmatrix} \qquad h_{\alpha_2} = \begin{pmatrix} 0 & & \\ & 1 & \\ & & -1 \end{pmatrix}$$

Let V be the defining representation of $\mathfrak{g}$ with standard basis $\{e_1, e_2, e_3\}$. We look for $\lambda_i \in \mathfrak{t}^*$ such that $V = \bigoplus_i V_{\lambda_i}$ with basis for each V_{λ_i} given by:

$$\begin{array}{ccc} h_{\alpha_1} e_1 = e_1, & h_{\alpha_2} e_2 = -e_2 & h_{\alpha_1} e_3 = 0 \\ h_{\alpha_2} e_1 = 0 & h_{\alpha_1} e_2 = e_2 & h_{\alpha_2} e_3 = -e_3 \end{array}$$

So let

$$\begin{aligned} \lambda_1(h_{\alpha_1}) = 1,\ \lambda_1(h_{\alpha_2}) = 0 &\Rightarrow V_{\lambda_1} = \langle e_1 \rangle \\ \lambda_2(h_{\alpha_1}) = -1,\ \lambda_2(h_{\alpha_2}) = 1 &\Rightarrow V_{\lambda_1} = \langle e_2 \rangle \\ \lambda_3(h_{\alpha_1}) = 0,\ \lambda_3(h_{\alpha_2}) = -1 &\Rightarrow V_{\lambda_3} = \langle e_3 \rangle \end{aligned}$$

Notice $\lambda_1 = \omega_1$, $\lambda_2 = -\omega_1 + \omega_2$, $\lambda_3 = -\omega_2$. on the diagram on page 81. Check e_1 is a HWV ($e_\alpha e_1 \in V_{\lambda+\alpha}$ etc.) [Fulton-Harris, Lectures 12,13].

17 The PBW theorem

Lemma 17.1.

1. *V has HWV*
2. *If $v \in V_\lambda$ is a highest weight vector, then λ must be dominant.*

Proof.

1. Choose any non-zero element $0 \neq v_0 \in V_\lambda$ for some λ. If it's a HWV then we're done, otherwise, there must be a positive root α such that $e_\alpha v_0 \neq 0$. Now let
$$k' = \max\{k \,:\, e_\alpha^k v_0 \neq 0\}$$
Let $v_1 = e_\alpha^k v_0 \in V_{\lambda + k'\alpha}$. Then repeat this argument, replacing v_0 by v_1, and keep going. This process *must* end, since V is finite dimensional, and each v_i is in a distinct weight-space V_μ, as we always add on a positive root.
2. For $\alpha \in \Phi^+$, need to show $\langle \lambda, \alpha^\vee \rangle \in \mathbb{Z}_{\geqslant 0}$. Consider $\mathfrak{m}_\alpha = \langle e_\alpha, h_\alpha, e_{-\alpha} \rangle$ acting on V.
$$e_\alpha v = 0, \qquad h_\alpha v = \lambda(h_\alpha) v$$
So v is a HWV for any $\mathfrak{m}_\alpha \cong \mathfrak{sl}_2$ acting on V, hece $\lambda(h_\alpha) \in \mathbb{Z}_{\geqslant 0}$.

□

Next, we show that there is a correspondence:

$$\{\text{ f.d. irreducible representations of } \mathfrak{g}\} \leftrightarrow \{\text{ dominant weights}\}$$

Universal enveloping algebra

For now, k may be any field. We associate to each Lie algebra, $\mathfrak{g}$ which is defined over k, an associative algebra with a 1 (infinite-dimensional in general (the algebra, not the one)) which is generated "as freely as possible" by $\mathfrak{g}$ subject to the commutation relations in $\mathfrak{g}$.

Definition 17.2. Let V be a vector space over the base field k. Define the tensor algebra of V:

$$\mathscr{T} := k \bigoplus V \bigoplus V \otimes V \bigoplus V \otimes V \otimes V \bigoplus \cdots$$

Or we can write this as

$$\bigoplus_{n \geqslant 0} V^{\otimes n} = \bigoplus_{j \geqslant 0} T^j V$$

(with $T^0 V = k$), with an associative product defined on homegeneous generators of $\mathscr{T}(V)$ by

$$v_1 \otimes \ldots v_m \cdot u_1 \otimes \cdots \otimes u_n = v_1 \otimes \ldots v_m \otimes u_1 \otimes \cdots \otimes u_n.$$

Definition (Symmetric algebra)**.** The **symmetric algebra** on V is $\mathrm{Sym}(V) = S(V) := \mathscr{T}(V)/I$ where I is the 2-sided ideal generated by $\{x \otimes y - y \otimes x \, : \, x, y \in V\}$. Notice that

- $S(V) = \bigoplus_{n \geqslant 0} S^n V$
- Can identify $S(V)$ with $k[V]$, the algebra of polynomials on V.
- $\mathscr{T}(V)$ and $S(V)$ are graded rings.

Definition 17.3 (Universal enveloping algebra)**.** Given an arbitrary Lie algebra $\mathfrak{g}$ (really arbitrary, possibly infinite dimensional), then the **universal enveloping algebra** $\mathscr{U}(\mathfrak{g})$ is the associative k-algebra

$$\mathscr{T}(\mathfrak{g})/J$$

where J is the 2-sided ideal generated by

$$\{x \otimes y - y \otimes x - [x, y] \, : \, x, y \in \mathfrak{g}\}$$

Some facts/exercises:

- We often write $x \otimes y$ as xy

- If V is a representation of $\mathfrak{g}$, then V is a $\mathscr{U}(\mathfrak{g})$-module:

$$(x_1 \otimes \dots x_n)v = x_1(\dots x_n(v)).$$

This is well-defined, because $(x \otimes y - y \otimes x)v = (xy - yx)v = [x, y]v$.

- If V is a finite dimensional $\mathbb{C}$-representation of $\mathfrak{g}$, i.e. $\mathfrak{sl}_2$, recall that we defined the Casimir element

$$\Omega = ef + fe + \frac{1}{2}h^2 \in \mathfrak{gl}(V).$$

Ω is naturally an element of $\mathscr{U}(\mathfrak{g})$, independent of V. In general, if $\mathfrak{g}$ is a semisimple complex Lie algebra, with basis $\{x_1, \dots, x_n\}$, then let $\{y_1, \dots, y_n\}$ be the dual basis with respect to the Killing form. Then

$$\Omega = \sum_{i=1}^{n} x_i y_i$$

which is in $\mathscr{U}(\mathfrak{g})$ by making the appropriate identifications. This is the Casimir element of (7.19). It's central in $\mathscr{U}(\mathfrak{g})$, just as it was central in $\mathfrak{sl}_2$.

- $\mathscr{U}(\mathfrak{g})$ is not graded, e.g. $\mathfrak{g} \otimes \mathfrak{g}$ is not closed under addition. For example

$$x \otimes y - y \otimes x - [x, y]$$

is not homogeneous, but is equal to 0. But it *does* have a filtration: Let

$$U_n = \text{ image of } \bigoplus_{i=0}^{n} \mathfrak{g}^{\otimes i}$$

in $scrU(\mathfrak{g})$. Then $U_n U_m \subseteq U_{n+m}$.

Exercise: If $x \in U_n$ and $y \in U_m$, then $xy - yx \in U_{m+n-1}$. Write

$$x = \sum_{i=1}^{p} \lambda_i x_1^i \dots x_n^i, \; y = \sum_{j=1}^{q} \mu_j y_1^j \dots y_m^j$$

Definition 17.4 (Graded algebra associated to a filtration)**.** Given any filtration

$$\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \dots,$$

call $\operatorname{gr} \mathcal{F} = \oplus_i \mathcal{F}_i / \mathcal{F}_{i-1}$ the **associated graded algebra**. In our case

$$\operatorname{gr} \mathscr{U}(\mathfrak{g}) = U_0 \oplus (\bigoplus_{n \geqslant 1} U_n / U_{n-1}).$$

Theorem 17.5 (PBW Theorem)**.** *There is an algebra isomorphism*

1. $S(\mathfrak{g}) \xrightarrow{\sim} \mathrm{gr}(\mathscr{U}(\mathfrak{g}))$
2. *Equivalently, if* $\{x_1, \ldots, x_n\}$ *is a basis for* $\mathfrak{g}$*, then*

$$\{x_1^{k_1}, \ldots, x_n^{k_n} \,:\, k_i \in \mathbb{Z}_{\geqslant 0}\}$$

is a basis for the universal enveloping algebra $\mathscr{U}(\mathfrak{g})$*. In particular* $\mathfrak{g} \hookrightarrow \mathscr{U}(\mathfrak{g})$*.*

Proof. Quite involved. Statement 1 is done in [Humphreys, 17.4]. The sketch is that, we have a map

$$\mathfrak{g} \xrightarrow{\iota} U_n \xrightarrow{/U_{n-1}} U_n/U_{n-1}$$

which gives a map from the tensor algebra to the associated graded algebra, which (by the exercise just before proposition 17.4) factors through the symmetric algebra $S(\mathfrak{g})$. So we have a commutative diagram[17]:

$$\begin{array}{ccc} \mathscr{T}(\mathfrak{g}) & \longrightarrow & \mathrm{gr}(\mathscr{U}) \\ & \searrow & \uparrow \\ & & S(\mathfrak{g}) \end{array}$$

This map is surjective (which is not hard to show). It remains to show that the map is injective (which *is* hard to show).

To prove 2, we have a basis for $S(\mathfrak{g})$, which gives a basis for $\mathrm{gr}(\mathscr{U})$. And this, in turn, gives a basis for $\mathscr{U}(\mathfrak{g})$. This is proved in [Humphreys, 17.3, Corollary C]. □

Lemma 17.6. *Suppose* V *is a representation of* $\mathfrak{g}$ *and* $v \in V$*. Then the minimal subrepresentation of* V *which contains* v *is*

$$\mathscr{U}(\mathfrak{g})v = \{uv : u \in \mathscr{U}(\mathfrak{g})\}.$$

Proof. $\mathscr{U}(\mathfrak{g})v$ contains:

- The elements $x_i v \ \forall x_i \in \mathfrak{g}$.

[17] as every Part III course must have.

- The scalar multiples of the above.
- The sums of the above.

□

Example. Let V be a $\mathbb{C}$-space, with basis $\{v_0, \dots, v_n\}$. Define an $\mathfrak{sl}_2$-action on V by:

$$e.v_0 = 0, \qquad h.v_0 = 0, \qquad f.v_i = v_{i+1}.$$

Then we claim that v_0 and v_1 are HWVs for the $\mathfrak{sl}_2$-action. So we require

$$ev_j = 0 \quad j = 0, 1$$

When $j = 0$ this is done by assumption.

When $j = 1$, we have

$$\begin{aligned} ev_1 &= efv_0 \\ &= ([ef] + fe)v_0 \\ &= [ef]v_0 \\ &= hv_0 \\ &= 0 \end{aligned}$$

We also require that $\langle v_0 \rangle$ and $\langle v_1 \rangle$ require their own images under h (the weight-space condition for HWVs).

For v_0 this is done, since it is clear.

For v_1:

$$\begin{aligned} hv_1 &= hfv_0 \\ &= ([hf] + fh)v_0 \\ &= [hf]v_0 \qquad\qquad\qquad\qquad = -2fv_0 = -2v_1 \end{aligned}$$

so $v_1 \in V_{-2}$ is a HWV. Also note that

$$\langle v_1, v_2, \dots, v_n \rangle \cong W$$

is a subrepresentation of V, and $V/W \cong V(0)$. Since it is a 1-dimensional representation. More generally, if $V^{(n)}$ is a space with basis $\{v_0, v_1 \dots\}$ with $\mathfrak{sl}_2$ action given by

$$ev_0 = 0, \qquad hv_0 = nv_0, \qquad fv_i = v_{i+1}.$$

then v_{n+1} is a HWV, and $W_n = \langle v_{n+1}, v_{n+2}, \dots \rangle$ has $V^{(n)}/W_n \cong V(n)$.

18 Highest weight and Verma modules

In this chapter, we take $\mathfrak{g}$ a Lie algebra, $\mathfrak{t}$ a CSA, Φ the set of roots of $\mathfrak{g}$ with respect to $\mathfrak{t}$, and $\Delta = \{\alpha_1, \ldots, \alpha_l\}$ root basis. If V is a representation of $\mathfrak{g}$, then

$$V_\lambda = \{v \in V \,:\, tv = \lambda(t)v \ \forall t \in \mathfrak{t}\},$$

as in (16.5).

Remark.

1. V_λ makes sense even if V is infinite dimensional.
2. The definition of HWV also makes sense if V is infinite dimensional.
3. If $e_\alpha \in \mathfrak{g}_\alpha$ is non-zero, then $e_\alpha V_\lambda \subseteq V_{\lambda+\alpha}$ (even if V is infinite-dimensional).

Definition 18.1 (Highest weight module)**.** A representation V of $\mathfrak{g}$ is a **highest weight module** if V contains a HWV, v, such that $V = \mathscr{U}(\mathfrak{g})v$.

Examples 18.2.

- Any finite dimensional irreducible representation V of $\mathfrak{g}$ is a highest weight module, since V must contain a highest weight vector v (17.1), and (17.6) implies that $\mathscr{U}(\mathfrak{g})$ is a subrepresentation of V, therefore $\mathscr{U}(\mathfrak{g}) = V$ since V is irreducible.
- In the above example, (end of §17), v_0 is a highest weight vector, $v_i = f^i v_0$, $V = \mathscr{U}(\mathfrak{g})v_0$ is a highest weight module.

Remark.

1. Not every highest weight module is irreducible.
2. If V is an infinite dimensional highest weight module, and $v \in V_\lambda$ a HWV, λ is not necessarily dominant (recall that it is dominant in the finite dimensional case ((17.1)2).)

Notation:

We will use η^+ to denote

$$\eta^+ = \sum_{\alpha \in \Phi^+} \mathfrak{g}_\alpha,$$

and η^- to denote

$$\eta^+ \sum_{\alpha \in \Phi^-} \mathfrak{g}_\alpha.$$

So we have

$$\mathfrak{g} = \eta^+ \oplus \mathfrak{t} \oplus \eta^-.$$

Lemma 18.3. *Suppose V is a highest weight module with HWV v such that $V = \mathscr{U}(\mathfrak{g})v$. Then $V = \mathscr{U}(\eta^-)v$.*

Proof. Choose a basis $\{x_1, \dots x_n\}$ of η^-, a basis $\{t_1, \dots, t_l\}$ of $\mathfrak{t}$, and $\{y_1, \dots, y_n\}$ of η^+. Then

$$\mathscr{U}(\mathfrak{g}) = \mathscr{U}(\eta^-) \oplus \mathscr{U}(\mathfrak{t}) \oplus \mathscr{U}(\eta^+),$$

and

$$\mathscr{U}(\mathfrak{g})v = \langle x_1^{k_1} \dots x_n^{k_n} t_1^{m_1}, \dots t_l^{k_l} y_1^{r_1} \dots y_n^{r_n} \rangle$$

But $y_i v = 0$ for all i, and $t_i v = \langle v \rangle$. Therefore

$$\mathscr{U}(\mathfrak{g})v = \mathscr{U}(\eta^-)v.$$

Proposition 18.4. *Let V be a highest weight module with highest weight vector given by $v_\lambda \in V_\lambda$ such that $V = \mathscr{U}(\mathfrak{g})v_\lambda$. Then*

1. *$V = \bigoplus_{\mu \in D(\lambda)} V_\mu$ where $D(\lambda) = \{\lambda - \sum_{i=1}^{l} k_i \alpha_i : k_i \in \mathbb{Z}_{\geqslant 0}\}$, where $\alpha_i \in \Delta$.*
2. *Any submodule of V is a direct sum of weight spaces.*
3. *The dimension, $\dim V_\lambda = 1$, and any other V_μ is finite dimensional.*
4. *V is irreducible iff every HWV lies in V_λ.*
5. *V contains some maximal (proper) subrepresentation.*

Proof.

1. By the PBW theorem, we can write $V = \mathscr{U}(\eta^-)v_\lambda$, therefore take expressions of the form

   $$e_{-\beta_2} e_{-\beta_2} \dots e_{-\beta_r} v_\lambda$$

   where $\beta_i \in \Phi^+$, $r \in \mathbb{Z}_{\geqslant 0}$ and $e_{-\beta_i} \in \mathfrak{g}_{-\beta_i}$. Expressions of this form span V. The weight of such an expression is $\lambda - \beta_1 - \dots - \beta_r$, so the generators live in $V_{\lambda - \sum \beta_i}$ where ($\beta_i \in \Phi^+$). So (1) and also (3) must hold, as given some μ, there exists only a finite number of ways to write μ as $\lambda - \sum \beta_i$ for $\beta_i \in \Phi^+$.

2. Let W be a submodule of V. Write $w \in W$ as a sum of vectors $v_i \in V_{\mu_i}$, each belonging to distinct weight spaces. We need to show that all of the v_i lie in W. Suppose they don't. Then we can choose a w such that

   $$w = \sum_{k=1}^{n} v_k$$

with n minimal, $n > 1$. In particular, no v_i belongs to W. Find $t \in \mathfrak{t}$ such that $\mu_1(t) \neq \mu_2(t)$. Then

$$tw = \sum \mu_i(t) v_i \in W.$$

Then also

$$(t - \mu_1(t).1)w = (\mu_2(t) - \mu_1(t))v_2 + \cdots + (\mu_n(t) - \mu_1(t))v_n \in W.$$

By the choice of w with n minimal forces $v_2 \in W$.

4. Suppose that V has HWV $v_\mu \in V_\mu$ where $\mu \neq \lambda$. Then $\mathscr{U}(\mathfrak{g})v_\mu$ is a subrepresentation of V, and $v_\lambda \notin \mathscr{U}(\mathfrak{g})v_\mu$. (Reason: weights for $\mathscr{U}(\mathfrak{g})v_\mu$ are of the form $\mu - \sum k_i\alpha_i$ for $k_i \in \mathbb{Z}_{\geqslant 0}$, so in particular, $\mathscr{U}(\mathfrak{g})v_\mu$ is a non-trivial proper subrepresentation).

   Conversely, suppose we can find $U \subseteq V$ a proper subrepresentation of V, then by (2), we can write U as a direct sum of weight spaces, V_μ. Choose μ to be such that $\mu = \lambda - \sum k_i\alpha_i$ where, as usual $k_i \in \mathbb{Z}_{\geqslant 0}$ such that $v_\mu \in U$ and $\sum k_i$ is minimal. Let $v_\mu \in V_\mu$ be non-zero, $\alpha \in \Phi^+$, and $e_\alpha \in \mathfrak{g}_\alpha$. Then

$$e_\alpha v_\mu \in V_{\mu+\alpha} \cap U = 0,$$

   by the choice of U. Hence v_μ is a HWV for V

5. Left as exercise. Take a proper submodule, which lies in the sum of the weight spaces other than λ, and take then sum of all such submodules, and this is proper. (In notes).

□

Exercises:

1. Show that V is indecomposable (as a $\mathfrak{g}$-module).
2. Show every non-zero homomorphic image of V is also a HWM.

18.1 Towards Verma Modules!

Let Φ, $\alpha \in \Delta$, λ is dominant., then $\lambda - \alpha$ is also dominant. In particular, $(\lambda - \alpha, \lambda - \alpha) \leqslant (\lambda, \lambda)$,

$$\begin{aligned}(\lambda - \alpha, \lambda - \alpha) &= (\lambda, \lambda) - (\alpha, \lambda) - (\lambda - \alpha, \alpha). \\ &= (\lambda, \lambda) - \text{something positive}\end{aligned}$$

which can be determined as a result of the fact that λ is dominant.

Definition 18.5 (highest weight of a HWM)**.** If V is a HWM with $v \in V_\lambda$ and $V = \mathscr{U}(\mathfrak{g})v$, where v is a HWV, we say that V is of **highest weight** λ.

Let $\mathfrak{t}$ be a CSA of $\mathfrak{g}$, Φ the corresponding root system, Δ a base, Φ^+ the set of positive roots w.r.t. Δ. Choose basis $h_1, \dots, h_l$ of $\mathfrak{t}$ such that $h_i = h_{\alpha_i}$ for $\alpha_i \in \Delta$. For $\lambda \in \mathfrak{t}^*$, let $J(\lambda)$ be a left-ideal of the universal enveloping algebra $\mathscr{U}(\mathfrak{g})$ generated by e_α for $\alpha \in \Phi^+$, and also combinations $h_i - \lambda(h_i) \cdot 1$ for $1 \leqslant i \leqslant l$. I.e. $J(\lambda)$ comprises elements

$$\sum u_\alpha e_\alpha + \sum y_i(h_i - \lambda(h_i) \cdot 1)$$

where $u_\alpha, y_i \in \mathscr{U}(\mathfrak{g})$. $J(\lambda)$ is a left-module for $\mathscr{U}(\mathfrak{g})$.

Definition 18.6 (Verma Module associated to λ)**.** Let $M(\lambda)$ be given as

$$M(\lambda) := \mathscr{U}(\mathfrak{g})/J(\lambda)$$

Then $M(\lambda)$ is a $\mathscr{U}(\mathfrak{g})$-module with action $u \cdot (v + J(\lambda)) = u \cdot v + J(\lambda)$. We say that $M(\lambda)$ is the **Verma Module associated to** $\lambda \in \mathfrak{t}^*$.

Proposition 18.7.

1. *This is HWM of highest weight λ.*
2. *$M(\lambda)$ is "universal" meaning that if $m_\lambda \in M(\lambda)_\lambda$ is a HWV and V is a HWM of highest weight λ, and highest weight vector $v_\lambda \in V_\lambda$ then there exists a unique $\mathfrak{g}$-equivariant linear map:*

$$\begin{aligned} M(\lambda) &\to V \\ m_\lambda &\mapsto v_\lambda \end{aligned}$$

Proof.

1. Let $m_\lambda = 1 + J(\lambda) \in M(\lambda)$ (a generator as a $\mathscr{U}(\mathfrak{g})$-module). Then

   $$h_i m_\lambda = h_i + J(\lambda) = \lambda(h_i) \cdot 1 + J(\lambda)$$

   since $h_i - \lambda(h_i) \in J(\lambda)$.

   And if $\alpha \in \Phi^+$ and $e_\alpha \in \mathfrak{g}_\alpha$, then $e_\alpha m_\lambda = e_\alpha + J(\lambda) = J(\lambda) = 0 + J(\lambda) \in M(\lambda)$, and m_λ is a HWV of weight λ, since it is annihilated by e_α. Clearly $M(\lambda) = \mathscr{U}(\mathfrak{g})m_\lambda$, so any other HWV is a scalar multiple of this one (18.4).
2. By PBW theorem, if $\Phi^+ = \{\beta_1, \dots, \beta_r\}$, then $e_{-\beta_1} \cdots e_{-\beta_r} \cdot m_\lambda$ $(r \in \mathbb{Z}_{\geqslant 0})$, is a basis for $M(\lambda)$. Define

   $$\begin{gathered} \varphi : M(\lambda) \to V \\ e_{-\beta_1} \cdots e_{-\beta_r} \cdot m_\lambda \mapsto e_{-\beta_1} \cdots e_{-\beta_r} v_\lambda \end{gathered}$$

   and check that this is $\mathfrak{g}$-equivariant.

Lemma 18.8. *Given $\lambda \in \mathfrak{t}^*$, there is a unique irreducible highest weight module with highest weight λ, call it $V(\lambda)$.*

Proof. You know that $M(\lambda)$ has unique maximal (proper) submodule (by 18.4) called J. Then $M(\lambda)/J$ is irreducible. Uniqueness comes from the universal property. □

Examples.

1. In the example at the end of §17, $V = M(0)$, $J = \langle v_1, v_2, \dots \rangle$, $M(0)/J \cong V(0)$, the trivial $\mathfrak{sl}_2$-representation.
2. [EW, Ex 15.12] give an example of irreducible Verma modules for $\mathfrak{g} = \mathfrak{sl}_2(\mathbb{C})$. This shows $\mathfrak{g}$ has infinite-dimensional irreducible representations.

Remark.

1. Verma modules are building blocks for "Category $\mathscr{O}$." Although each $M(\lambda)$ is infinite-dimensional, when viewed as $\mathscr{U}(\mathfrak{g})$-modules, they have "finite length," i.e. there are submodules

$$0 \subseteq M_0 \subseteq M_1 \subseteq \cdots \subseteq M_r = M(\lambda)$$

   such that M_i/M_{i-1} are irreducible.
2. In 1985 Drinfield and Jimbo (independently) defined **quantum groups** by "deforming" the universal enveloping algebras of Lie algebras. They have numerous applications in theoretical physics, knot theory, and the representation theory of algebraic groups.

Theorem 18.9. *$V = V(\lambda)$ is a finite dimensional, irreducible $\mathfrak{g}$-module if and only if $\lambda \in X^+$.*

Proof.

(⇒) If V is finite dimensional, then for each simple root α_i, let $\mathfrak{m}_{\alpha_i}$ be the corresponding copy of $\mathfrak{sl}_2(\mathbb{C})$ in $\mathfrak{g}$. Then V is also a finite dimensional module for $\mathfrak{m}_{\alpha_i}$, and a HWV for $\mathfrak{g}$ is also a HWV for $\mathfrak{m}_{\alpha_i}$. In particular, since there is a highest weight vector of weight λ, the wight for the CSA $A_i \subseteq \mathfrak{m}_{\alpha_i}$ is determined by $\lambda(h_i)$ since

$$h_{\alpha_i} v := h_i(v) = \lambda(h_i)v = \langle \lambda, \alpha_i^\vee \rangle v$$

This forces $\lambda(h_i) \in \mathbb{Z}_{\geqslant 0}$.

($\Leftarrow$) $V(\lambda)$ is a direct sum of finite dimensional weight spaces (since $M(\lambda)$ is). The idea is to show that $\Pi(\lambda) = \{\mu : V(\lambda)_\mu \neq 0\}$ (a set of weights) is finite.

Have $\Delta = \{\alpha_1, \ldots, \alpha_l\}$ a set of simple roots. For each i, $\{x_i, y_i, h_i\}$ is an $\mathfrak{m}_{\alpha_i}$-triple. We need that, for $k \geqslant 0$ (in $\mathscr{U}(\mathfrak{g})$):

(i) $[x_j, y_i^{k+1}] = 0$ for $i \neq j$

(ii) $[x_i, y_i^{k+1}] = -(k+1)y_i^k(k \cdot 1 - h_i)$.

these are proved in [Humphreys, 21.2]

1. First of all, we show that $V(\lambda)$ contains a non-zero, finite dimensional $\mathfrak{m}_{\alpha_i}$-module for each i.

   Let v be a HWV in $V(\lambda)$. As $\lambda \in X^+$, $n_i = \langle \lambda, \alpha_i^\vee \rangle = \lambda(h_i) \in \mathbb{Z}_{\geqslant 0}$. Then we claim $u = y_i^{n_i+1}v = 0$.

   Claim (i) above implies that

   $$x_j u = y_i^{n_i+1}(x_j v) + [x_j y_i^{n_i+1}] = 0$$

   for $(i \neq j)$.

   And claim (ii) above implies that

   $$xiu = y_i^{n_i+1}(x_i v) + [x_i y_i^{n_i+1}]v = -(n_i+1)y_i^{n_i}(n_i - h_i)v = 0,$$

   as $h_i v = n_i v$.

   If $u \neq 0$, u would be a HWV of weight $\lambda - (n_i+1)\alpha_i < \lambda$. Contradiction, as highest weight is unique, so

   $$W = \langle v, y_i v, \ldots, y_i^{n_i+1} v \rangle$$

   is a non-zero finite dimensional $\mathfrak{m}_{\alpha_i}$-subrepresentation of $V(\lambda)$.

   [Check: $h_i W \subseteq W$, $y_i W \subseteq W$ clear, $x_i W \subseteq W$ needs (i) above.]

2. For each i, $V(\lambda)$ is the sum of all finite-dimensional $\mathfrak{m}_{\alpha_i}$-subrepresentations contained in it.

   Let W = sum of all finite dimensional $\mathfrak{m}_{\alpha_i}$-modules contained in $V(\lambda)$. Show W i a $\mathfrak{g}$-submodule of $V(\lambda)$. With (1) (says $W \neq 0$), this forces $W = V(\lambda)$. For $x \in \mathfrak{g}$, $w \in W$, need $xw \in W$. Since $w \in W$, we must have $w \in W'$ for some finite dimensional $\mathfrak{m}_{\alpha_i}$module W'. Let $x' = \sum_{\beta \in \Phi \cup \{0\}} x_\beta$, $x_\beta \in \mathfrak{g}_\beta$, then $xw \in \operatorname{span}_\beta\{x_\beta W'\} = W''$. W'' is finite dimensional, and clearly it is $\mathfrak{m}_{\alpha_i}$-invariant ($x_i W'' \subseteq \operatorname{span}_\beta\{x_i x_\beta W'\}$), and so

   $$x_i x_\beta W' = x_\beta(x_i W') + [x_i, x_\beta]W' \subseteq x_\beta W' + \mathfrak{g}_{\alpha_i+\beta} W' \subseteq W''$$

Similarly, repeat for y_i, h_i. So

$$xw \in W'' \subseteq W.$$

Now show Weyl group acts on $\Pi(\lambda)$ by permutations. If so, then $\Pi(\lambda)$ decomposes as disjoint union of Weyl-orbits. This will mean it's enough to show there are only finitely many orbits: use this together with the fact that the Weyl group is finite to get $\Pi(\lambda)$ finite.

3. $\mu \in \Pi(\lambda) \Rightarrow w_i\mu \in \Pi(\lambda)$, where w_i is the reflection corresponding to α_i. Also $\dim V(\lambda)_\mu = \dim V(\lambda)_{w_i\mu}$. Since $V(\lambda)_\mu$ is finite dimensional, there is, by (2), a finite dimensional $\mathfrak{m}_{\alpha_i}$-module U containing $V(\lambda)_\mu$.

Lecture 13

For $0 \neq w \in V(\lambda)_\mu$, we have $h_{\alpha_i} w := h_i w = \mu(h_i)w$. But $w \in U$ is then a weight vector for h_i, so by $\mathfrak{sl}_2$-theory,

$$\mu(h_i) = \langle \mu, \alpha_i^\checkmark \rangle = \text{an integer, } m.$$

Therefore all weights are in $\mathbb{Z}$, as this holds for all i. As m occurs as a weight of U, so must $-m$, and $\dim U_m = \dim U_{-m}$.

If $m \in \mathbb{Z}_{\geqslant 0}$, then $y_i^m w \neq 0$, and lives in U_{-m}. But,

$$y_i^m w \in V(\lambda)_{\mu - \langle \mu, \alpha_i^\checkmark \rangle \alpha_i}$$

because "weights add", that is to say, $w_i\mu$ is also a weight. If $m \in \mathbb{Z}_{<0}$ just take $x_i^{-m} w$, and apply the same argument.

Now we need to show that the dimensions are equal. Suppose that $\{w_1, \ldots, w_r\}$ is a basis of $V(\lambda)_\mu$, then $\{w_1, \ldots, w_r\}$ is linearly independent in U_m. Therefore, applying y_i^m or x_i^{-m} as appropriate leaves the vectors linearly independent, and sends them to $V_{\mu - \langle \mu, \alpha_i^\checkmark \rangle \alpha_i} = V_{w_i\mu}$. So

$$\dim V(\lambda)_\mu \leqslant \dim V(\lambda) w_i \mu.$$

Applying the same procedure to $w_i\mu$ gives

$$\dim V(\lambda)_{w_i\mu} \leqslant \dim V(\lambda)_{w_i^2\mu}.$$

But w_i is a reflection, so $w_i^2 = e$.

4. For $\mu \in \Pi(\lambda)$, its Weyl orbit[18] $W\mu$ contains a dominant weight.

$W\mu$ is finite, therefore we can choose $\eta \in W\mu$ maximal with respect to $\leqslant$. Then η must be dominant, since supposing it weren't, we would have

$$\langle \eta, \alpha_i^\checkmark \in \mathbb{Z}_{<0} \rangle$$

[18]The orbit of the Weyl group acting on μ.

for some i. Then $w_i\eta \in W\mu$, $w_i\eta = \eta - \langle \eta, \alpha_i^\vee \rangle \alpha_i > \eta$, contradiction to the chocie of η.

5. The set $S = \{\eta : \eta \text{ dominant}, \eta \leqslant \lambda\}$ is finite.

   If $\eta \in S$, then $\lambda - \mu =$ sum of positive roots with $\mathbb{Z}_{\geqslant 0}$ coefficients. Hence η lies in a discrete set. Also $\lambda + \eta$ is dominant, therefore

   $$\langle \lambda + \eta, \alpha_i^\vee \rangle \geqslant 0 \quad \forall i,$$

   in particular

   $$(\lambda + \eta, \lambda - \eta) \geqslant 0 \Rightarrow (\lambda, \lambda) \geqslant (\eta, \eta),$$

   So η lies in a compact set, since it is in the closed ball of radius $\sqrt{(\lambda, \lambda)}$. Compact and discrete means finite, so S must be finite.

From (4), any W-orbit of $\Pi(\lambda)$ contains an element of S, and (5) implies that the number of orbits is finite.

□

So there is a bijection

$$\{\text{ dominant weights, } \lambda\} \leftrightarrow \{\text{f.d. irred representations } V(\lambda) \text{ of } \mathfrak{g}.\}$$

19 The Weyl character formula

Let $\mathfrak{g}$ be a semisimple Lie algebra, $\mathfrak{t}$ a CSA, $\Delta = \{\alpha_1, \ldots, \alpha_l\} \subseteq \Phi$ a choice of root basis inside the set of roots, X the weight lattice, W is the Weyl group.

Example. Define

$$\rho = \frac{1}{2} \sum_{\alpha \in \Phi^+} \alpha.$$

Compute $\langle \rho, \alpha_i^\vee \rangle$ for each $\alpha_i \in \Delta$ in types A_1, A_2, and B_2. For A_2, B_2, we can write $\Delta = \{\alpha_1, \alpha_2\}$. Then we claim that $\rho = \omega_1 + \omega_2$:

- A_1: $\{\alpha_1, -\alpha_1\}$, $\rho = \frac{\alpha}{2}$, $\langle \rho, \alpha^\vee \rangle = 1$.
- A_2: $\langle \rho, \alpha_1^\vee \rangle = \langle \omega_1 + \omega_2, \alpha_1^\vee \rangle = 1$
- B_2: $\langle \rho, \alpha_i^\vee \rangle = 1$.

Then the claim is that $\rho = \sum_{i=1}^{l} \omega_i$ (ω_i as defined in (16.3)).

It is enough to show that $\langle \rho, \alpha_j^\vee \rangle = 1$ for all $\alpha_j \in \Delta$. Now

$$w_{\alpha_j}(\rho) = \rho - \langle \rho, \alpha_j^\vee \rangle \alpha_j,$$

and also

$$w_{\alpha_j}(\rho) = w_{\alpha_j}\left(\frac{1}{2}\left(\sum_{\alpha\in\Phi^+\setminus\{\alpha_j\}}\alpha\right)+\frac{1}{2}\alpha_j\right),$$

and w_{α_j} permutes $\Phi^+ \setminus \{\alpha_j\}$, so this is

$$\frac{1}{2}\left(\sum_{\alpha\in\Phi^+\setminus\{\alpha_j\}}\right)-\frac{1}{2}\alpha_j = \rho - \alpha_j.$$

Comparing the two expressions, $\langle \rho, \alpha_j^\checkmark \rangle = 1$.

Recall from (18.9) that $\Pi(\lambda) = \{\mu : V(\lambda)_\mu \neq 0\}$. Possible questions:

- What is $\Pi(\lambda)$?
- What is $\dim V(\lambda)$?

Definition 19.1 (Ordering on X)**.** Partial ordering $\leqslant$ (some books write it $\preceq$) on X by

$$\mu \leqslant \lambda \Leftrightarrow \lambda - \mu = \sum_{\alpha_i\in\Delta} k_i\alpha_i \quad k_i \in \mathbb{Z}_{\geqslant 0}.$$

Note that

1. $\Pi(\lambda) \subseteq \{\mu : \mu \leqslant \lambda\}$
2. To determine $\Pi(\lambda)$ we need only to find the dominant weights in it.

Proposition 19.2. *Suppose λ, μ are both dominant, then*

$$\mu \in \Pi(\lambda) \Leftrightarrow \mu \leqslant \lambda.$$

Proof. Suppose $\mu \leqslant \lambda$. Then

$$\lambda - \mu = \sum_{\alpha\in\Phi^+} k_\alpha\alpha, \quad k_\alpha \in \mathbb{Z}_{\geqslant 0}.$$

We proceed by induction on $\sum k_\alpha$. We've already done the case where $\sum k_\alpha = 0$, since this is just saying that $\lambda = \mu$.

The next case, is that $\mu = \lambda - \alpha$ for $\alpha \in \Phi^+$, i.e. μ is one positive root away from λ. Then we compute the usual bracket:

$$\langle \mu, \alpha^\checkmark \rangle = \langle \lambda, \alpha^\checkmark \rangle - 2 \geqslant 0.$$

Therefore $\langle \lambda, \alpha^\vee \rangle \geqslant 2$. Now take $v_\lambda \in V_\lambda$ to be non-zero. Since $h_\alpha v_\lambda = n v_\lambda$, for some $n \geqslant 2$ and we know $e_{-\alpha} v_\lambda \neq 0$ (by usual $\mathfrak{sl}_2$ theory: it's in $V(\lambda)_{\lambda-\alpha} = V(\lambda)_\mu$).

Suppose we know the claim is true when $\sum k_\alpha = n - 1$. Now assume that $\sum k_\alpha = n$ so that

$$\mu = \lambda - \beta_1 - \beta_2 - \cdots - \beta_n.$$

There are two cases:

Case 1. $\langle \beta_i, \beta_j^\vee \rangle < 0$ for some i, j such that $i < j$ WLOG. Note that $\beta_i + \beta_j$ is a positive root (why?).

Therefore

$$\begin{aligned}\sum_{k=1}^{n} \beta_k &= \left(\sum_{k=1}^{i-1} \beta_k \right) + \left(\sum_{k=i+1}^{j-1} \beta_k \right) + \beta_i + \beta_j + \left(\sum_{k=j+1}^{n} \beta_k \right) \\ &= \sum_{k=1}^{n-1} \gamma_k\end{aligned}$$

where

$$\gamma_k = \begin{cases} \beta_k, & k \leqslant i-1 \\ \beta_{k+1}, & i \leqslant k \leqslant j-2 \\ \beta_{k+2}, & j-1 \leqslant k \leqslant n-2 \\ \beta_i + \beta_j, & k = n-1 \end{cases}$$

so $\gamma_k \in \Phi^+$ for all k, and by the inductive hypothesis, we're done.

Case 2. $\langle \beta_i, \beta_j^\vee \rangle \geqslant 0$ for all $i \neq j$. For this, we need to prove the following claim:

Claim.

$$\lambda - \sum_{i=1}^{r} \beta_i \in \Pi(\lambda) \quad \forall 1 \leqslant r \leqslant n.$$

Proved by induction on r, in the written notes.

Example. $\{\alpha_1, \alpha_2\}$ base of G_2[19] with α_1 short. Compute $\Pi(2\omega_1)$.
Recall that ω_1 is such that $\langle \omega_1, \alpha_1^\vee \rangle = 1$ and $\langle \omega_1, \alpha_2^\vee \rangle = 0$. The dominant weights in $\Pi(2\omega_1)$ are exactly the dominant μ which are less than ω_1, i.e. $\mu \prec 2\omega_1$. We have

$$2\omega_1 = 2(\alpha_2 + 2\alpha_1) = 2\alpha_2 + 4\alpha_1$$

and we have

$$\omega_2 = 3\alpha_1 + 2\alpha_2$$

[19] For a useful picture, check pages 58, 69 of Humphreys. Fulton and Harris have many many more calculations on pages 339-359.

So $\omega_2 \preceq 2\omega_1$. But $\omega_1 + \omega_2 \not\preceq 2\omega_1$.

So the dominant weights in $\Pi(2\omega_1)$ are $\{\omega_1, 2\omega_1, \omega_2, 0\}$. To get everything else in $\Pi(2\omega_1)$ we need to let the Weyl group act on these elements to determine their orbits. The Weyl conjugates[20] of ω_1 are the short roots, and the Weyl conjugates of ω_2 are all the long roots. So $\Pi(2\omega_1)$ is made up of all the short roots, twice of all the short roots, and all the long roots. This turns out to be

$$\Phi \cup \{2 \pm \omega_1, \pm 2\alpha_1, \pm 2(\alpha_1 + \alpha_2), 0\}$$

"This is a very typical exam question!"

Definition 19.3 ((Formal) character of V)**.** Let $\mathbb{Z}[X]$ be the free $\mathbb{Z}$ module with basis $\{e^\mu : \mu \in X\}$, with multiplication $e^\mu e^\lambda = e^{\mu+\lambda}$. This is a commutative ring when the operation is extended linearly, with identity $1 = e^0$.

Let V be a finite dimensional representation of $\mathfrak{g}$. Then the **(formal) character of** V is:

$$\mathrm{ch}(V) = \sum_{\mu \in X} (\dim V_\mu) e^\mu \in \mathbb{Z}[X]$$

Example.

1. $\mathfrak{g} = \mathfrak{sl}_2$, $X = \mathbb{Z}\frac{\alpha}{2}$. Put $z = e^{\alpha/2}$, then
$$\mathrm{ch}(V(n)) = z^n + \cdots + z^{-n}.$$
2. **Exercise:** Find the character of the adjoint representation of $\mathfrak{sl}_3$. (These are done in some detail in Grojnowski's notes.)

Recall (Ex. Sheet 3) that $l(w)$ is the minimal n such that w can be written as a product of n simple reflections w_{α_i}.

Definition (Sign of w)**.** For $w \in W$, we write the **sign of** w as:

$$\mathrm{sgn}(w) = (-1)^{l(w)}$$

Example. $\mathfrak{g} = \mathfrak{sl}_n$, then $\mathrm{sgn}(w) = \mathrm{sgn}(w')$ where w' is the image of w in S_n, as $S_n \cong W$ (by "14.4" [21]).

[20] This is what we call the result of some element when we apply the Weyl group to it.
[21] I think it's actually 11.4 but unsure.

Theorem 19.4 (Weyl Character Formula)**.** *If λ is a dominant weight and*

$$\rho = \frac{1}{2}\sum_{\alpha\in\Phi^+}\alpha = \sum_{j=1}^{n}\omega_j$$

then

$$\mathrm{ch}(V(\lambda)) = \frac{\sum\limits_{w\in W}\mathrm{sgn}(w)e^{w(\lambda+\rho)}}{e^{\rho}\left(\prod\limits_{\alpha\in\Phi^+}(1-e^{-\alpha})\right)}$$

Proof. Omitted. Can be read in Fulton and Harris, chapter 24; Grojnowski's notes, chapter 7; Humphreys, 24.3.

Corollary 19.5 (The Weyl denominator identity)**.**

$$e^{\rho}\prod_{\alpha\in\Phi^+}(1-e^{-\alpha}) = \sum_{w\in W}\mathrm{sgn}(w)e^{w\rho}.$$

Proof. Obvious, $\mathrm{ch}(V(0)) = 1$, so plug $\lambda = 0$ into the formula in (19.4). □

Corollary 19.6 (Weyl dimension formula)**.** *If λ is dominant, then*

$$\dim V(\lambda) = \frac{\prod\limits_{\alpha\in\Phi^+}\langle\lambda+\rho,\alpha^{\vee}\rangle}{\prod\limits_{\alpha\in\Phi^+}\langle\rho,\alpha^{\vee}\rangle} = \prod_{\alpha\in\Phi^+}\frac{(\lambda+\rho,\alpha)}{(\rho,\alpha)}.$$

Proof. By definition:

$$\mathrm{ch}(V(\lambda)) = \sum_{\alpha\in X^+}\dim V(\lambda)_{\mu}e^{\mu}.$$

We'd like to substitute $e^{\mu} = 1$ into (19.4) for any μ, but this could give us $\frac{0}{0}$. Instead, for $\mu \in X$, $p \in \mathbb{Z}[X]$, define a function:

$$F_{\mu} : \mathbb{R}_{\geqslant 0} \to \mathbb{R}$$
$$F_{\mu}(e^{\lambda})(q) = q^{-(\mu,\lambda)}$$

and since this is defined on the basis elements e^{λ}, we can extend this linearly. Note that F_{μ} is multiplicative and $F_{\mu}(p)$ is a continuous differentiable function

on $\mathbb{R}_{>0}$. Clearly $F_0(e^\lambda) = 1$, so $F_0(\mathrm{ch}(V(\lambda))) = \dim V(\lambda)$.

First, apply F_μ to (19.5), so for all $\mu \in X$ and all $q \in \mathbb{R}_{>0}$,

$$\begin{aligned} q^{-(\rho,\mu)} \prod_{\alpha\in\Phi^+} (1 - q^{(\alpha,\mu)}) &= \sum_{w\in W} \mathrm{sgn}(w) q^{-(w\rho,\mu)} \\ &= \sum_{w\in W} \mathrm{sgn}(w) q^{(\rho,w\mu)} \qquad (*) \end{aligned}$$

where we can move the w to the other side in the exponent of q since $\mathrm{sgn}(w) = \mathrm{sgn}(w^{-1})$ and $(x, wy) = (w^{-1}x, y)$.

Now apply F_p to (19.4) to obtain

$$F_p(\mathrm{ch}(V(\lambda))) = \frac{\sum_{w\in W} \mathrm{sgn}(w) q^{(-p,w(\lambda+\rho))}}{q^{(-p,\rho)} \prod_{\alpha\in\Phi^+} (1 - q^{(p,\alpha)})}$$

if $(p, \alpha) \neq 0$ for all $\alpha \in \Phi^+$. But recall, $(\rho, \alpha_i) = 1$[22] for all simple roots α_i. So $(\rho, \alpha) > 0$ for all $\alpha \in \Phi^+$. Now, using (*) along with $\mu = \lambda + \rho$ we have:

$$F_p(\mathrm{ch}(V(\lambda))) = \frac{q^{-(p,\lambda+\rho)} \prod_{\alpha\in\Phi^+} (1 - q^{(\alpha,\lambda+\rho)})}{q^{-(p,\rho)} \prod_{\alpha\in\Phi^+} (1 - q^{-(p(\alpha))})}.$$

Finally, note that:

$$F_p(\mathrm{ch}(V(\lambda)))(q) = \sum (\dim V(\lambda))_\mu q^{-(p,\mu)}$$

So taking the limit $q \to 1$, and using L'Hôpital's Rule, we get:

$$\dim V(\lambda) = \frac{\prod_{\alpha\in\Phi^+} (\lambda + \rho, \alpha)}{\prod_{\alpha\in\Phi^+} (\rho, \alpha)} = \frac{\prod_{\alpha\in\Phi^+} \langle \lambda + \rho, \alpha^\vee \rangle}{\prod_{\alpha\in\Phi^+} \langle \rho, \alpha^\vee \rangle}$$

Examples.

1. $\mathfrak{g} = \mathfrak{sl}_2$, $\omega_1 = \frac{1}{2}\alpha = \rho$ since there is only one root in the base. $X^+ = \{m\omega_1\}$ where $m \in \mathbb{N}_0$. Then for $\lambda \in X^+$, we have

   $$\dim V(\lambda) = \frac{(m\omega_1 + \rho, \alpha)}{(\omega_1, \alpha)} = \frac{(m+1)(\omega_1, \alpha)}{(\omega_1, \alpha)} = m + 1$$

   which we expect from standard $\mathfrak{sl}_2$ theory.

[22] Recall that ρ is half the sum of all the positive roots, also equal to the sum of the fundamental weights. It has the property that it always has inner product 1 with the simple roots (we calculated this right after the beginning of section 19).

2. $\mathfrak{g} = \mathfrak{sl}_3$ $\Phi^+ = \{\alpha, \beta, \alpha+\beta\}$. Let $\lambda = m_1\omega_1 + m_2\omega_2$. In this case, $\rho = \omega_1 + \omega_2 = \frac{1}{2}(\alpha + \beta + (\alpha+\beta))$. So, calculating $\dim(V(\lambda))$, we need to calculate the inner products of $\lambda + \rho$ with the positive roots, and ρ with the positive roots.

	α	β	$\alpha+\beta$
$(\lambda+\rho, -)$	m_1+1	m_2+1	m_1+m_2+2
$(\rho, -)$	1	1	2

The first row gives the numerators, and the second row gives the denominators, so putting them all together, we get

$$\begin{aligned}\prod \langle \lambda+\rho, \alpha^\vee \rangle =& \langle m_1\omega_1 + m_2\omega_2 + \alpha_1 + \alpha_2, \alpha_1^\vee \rangle \\ &\times \langle m_1\omega_1 + m_2\omega_2 + \alpha_1 + \alpha_2, \alpha_2^\vee \rangle \\ &\times \langle \dots, (\alpha_1+\alpha_2)^\vee \rangle\end{aligned}$$

Therefore $\dim V(\lambda) = \frac{(m_1+1)(m_2+1)(m_1+m_2+2)}{2}$

Exercise: Compute the dimensions of all the finite dimensional irreducible representations of B_2, G_2, [Humphreys, Page 140].

Example. $\mathfrak{g} = \mathfrak{sp}_4(\mathbb{C})$. Let α_1 be the short root, and recall that this is a 10-dimensional Lie algebra of type B_2.

1. Find $\dim(V(\lambda))$, $\lambda = a\omega_1 + b\omega_2 \in X^+$. Calculate $\langle \omega_1, \alpha_1^\vee \rangle = 1$, $\langle \omega_2 \alpha_2^\vee \rangle = 0, \dots$ Proceed with all 4 positive roots and the two ω_i, so 8 calculations in total. Then use this to calculate:

$$\prod_{\alpha \in \Phi^+} \langle \rho, \alpha^\vee \rangle = 6.$$

We have $\lambda + \rho = (a+1)\omega_1 + (b+1)\omega_2$, so

$$\prod_{\alpha \in \Phi^+} \langle \lambda+\rho, \alpha^\vee \rangle = (a+1)(b+1)(a+2b+3)(a+b+2).$$

Dividing these two expressions as in the dimension formula gives the formula for the dimension of $V(\lambda)$, i.e. the above expression divided by 6.

2. Let V be the defining representation (of dimension 4). Find the highest weight:

Well, $\dim(V(\omega_1)) = 4$, so if W is a non-trivial representation of $\mathfrak{sp}_4$ which is not isomorphic to $V(\omega_1)$, then by the dimension formula, you must have $\dim W > 4$. So $V \cong V(\omega_1)$. So the highest weight is ω_1.

3. Decompose $V \otimes V$ into irreducible subrepresentations.

   In order to get this decomposition, let $v \in V(\omega_1)$ be a HWV, then $v \otimes v$ is a HWV of $V \otimes V$, with weight obtained by summing the ω_1 weights of the v, so $v \otimes v$ has weight $2\omega_1$.

   Therefore $V(2(\omega_1))$ is a subrepresentation of $V \otimes V$. Now $2\omega_1 = 2\alpha_1 + \alpha_2$, so $V(2\omega_1)$ is the adjoint representation[23]. In particular, $V(2\omega_1)_{\omega_2}$ has dimension 1.

   Finally, take $\{v_\gamma : \gamma \in \Pi(\omega_1)\}$ a basis of weight vectors of V. Then $\{v_{\gamma_1} \otimes v_{\gamma_2} : \gamma_1, \gamma_2 \in \Pi(\omega_1)\}$ forms a basis of weight vectors for $V \otimes V$.

   Then, with a bit of work:

   - $(V \otimes V)_{\omega_1}$ has dimension 1.
   - $(V \otimes V)_{\omega_2}$ has dimension 2.
   - $(V \otimes V)_0$ has dimension 4.

   All the other weight spaces correspond to non-dominant weights. Also $V(2\omega_1)_{\omega_2}$ has dimension 1. $V(\omega_2)$ must be a subrepresentation of $V \otimes V$, but $\dim V(\omega_2) = 5$. BY counting dimensions, we have:

$$V \otimes V = V(2\omega_1) \oplus V(\omega_2) \oplus V(0)$$

   this agrees with counting dimensions, giving $16 = 10 + 5 + 1$.

[23] By considering the diagram for B_2 we can see this.